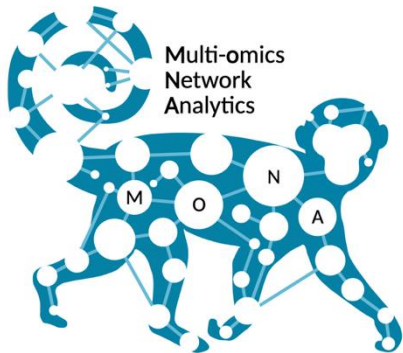


Informatic Platform and MoNA

From proteins to data: an introduction to Proteomics

Date: 03 June 2026



Marco Reverenna, PhD student

Henry Webel, Senior Data Scientist

Alberto Santos, Director Informatic Platform

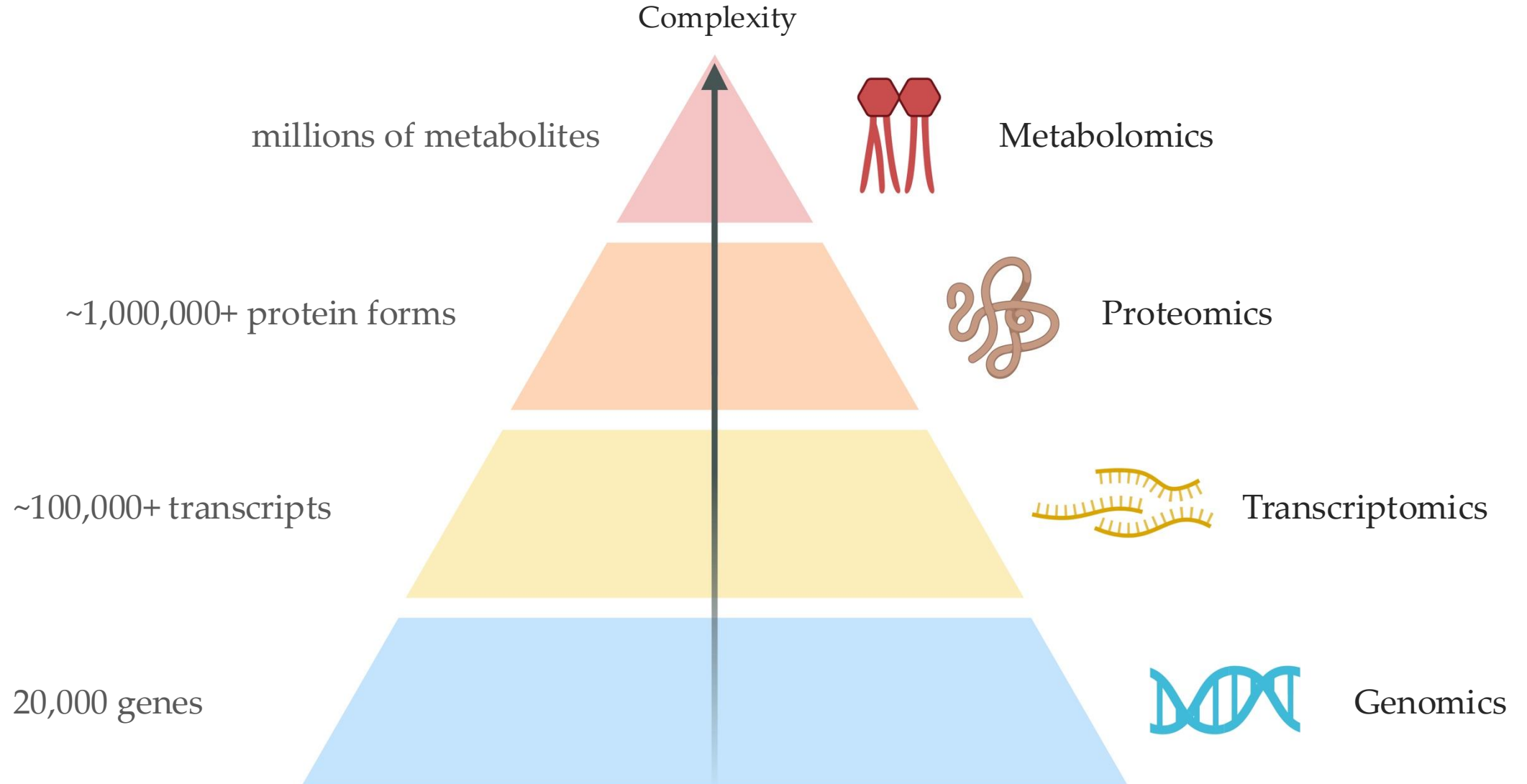
Alberto Pallejà, Team Lead Data Science Platform

Day plan!

Time	Topic	lecturer
09.00 - 09:10	Welcome, coffee, and setup	-
09:10 - 11.00	Introduction with overview of all the components	Marco Reverenna
11.00 - 11.15	break	-
11.15 - 12.15	Steps in data processing and running quantms hands-on	Henry Webel
12.15 - 13.00	Lunch (sandwiches are provided)	-
13.00 - 14.30	Steps in statistical analysis (lecture)	Alberto Santos
14.30 - 15.00	break	-
15.00 - 16.30	Steps in statistical analysis (Hands-On)	Henry Webel

Each 90mins session will have a small break of 5mins inbetween.

Omics sciences



Proteomics

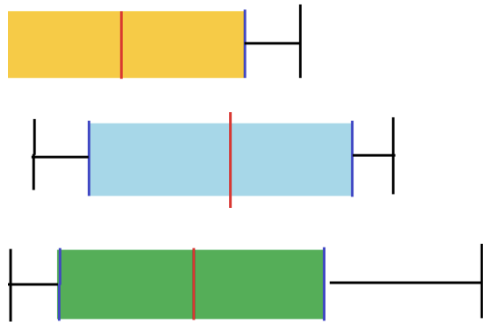
“Proteomics is the large-scale study of interaction, function, composition and structure of proteins and their cellular activities”

Identification

Which proteins are present?

Which proteins are uniquely expressed in a specific conditions?

What unannotated or novel peptides are hidden in the sample?



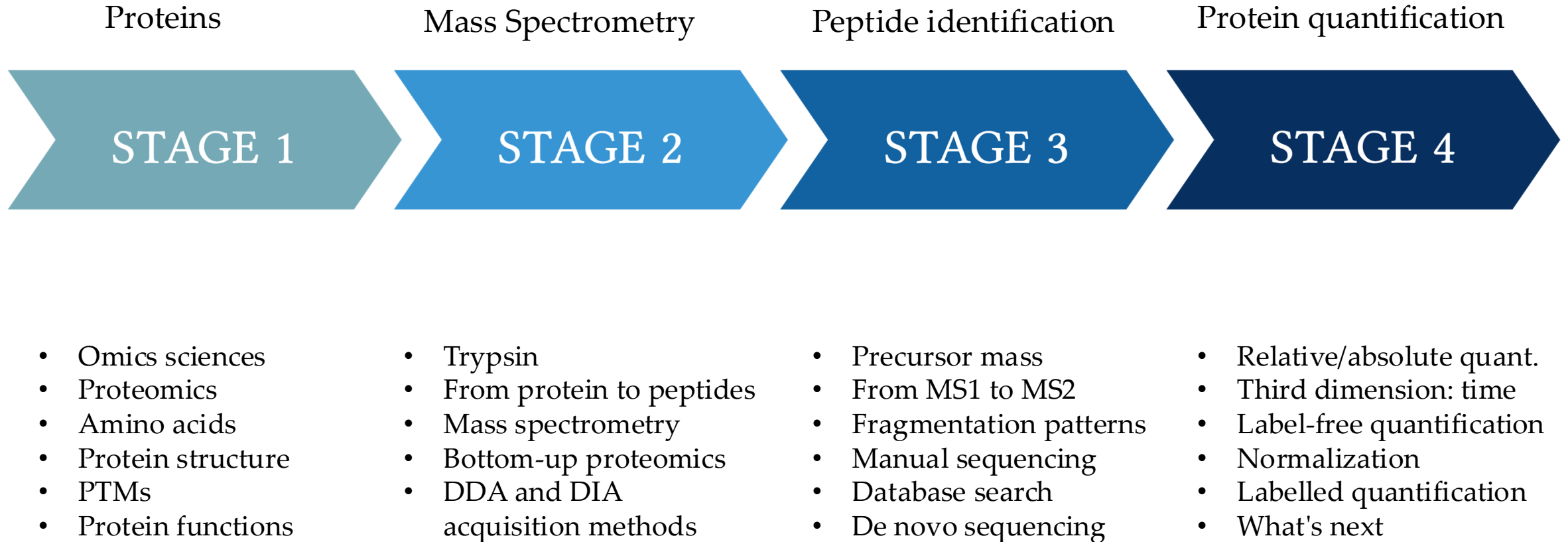
Quantification

How many copies of each protein do we have?

How do proteins abundance levels shift between conditions?

What the proteome changes in a specific environment?

Proteomics intro roadmap



From DNA to proteins...

DNA



Transcription



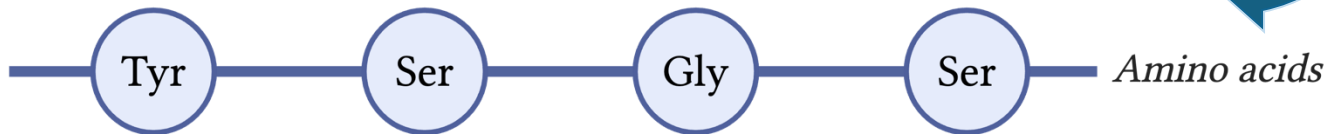
mRNA



Translation



Protein



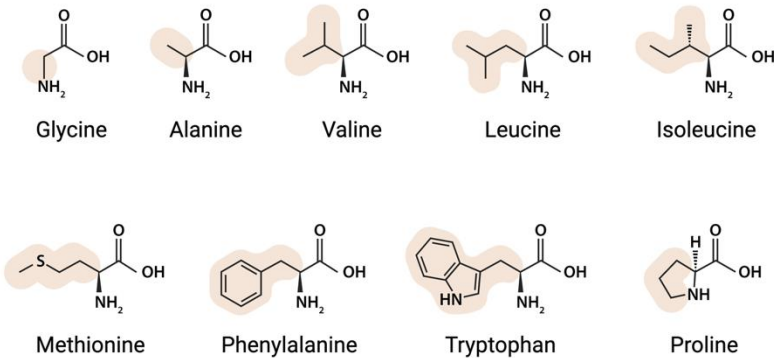
Unlike DNA, proteins are dynamic: their **abundance**, **modifications** and **activity** change continuously with the cell's state.

This is why **we cannot infer** the proteome from the genome.

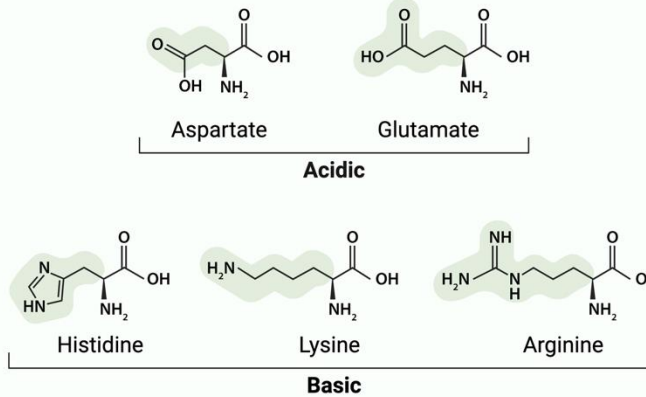
20 standard amino acids

Each amino acid has a unique residue mass and we can distinguish them in 3 main categories.

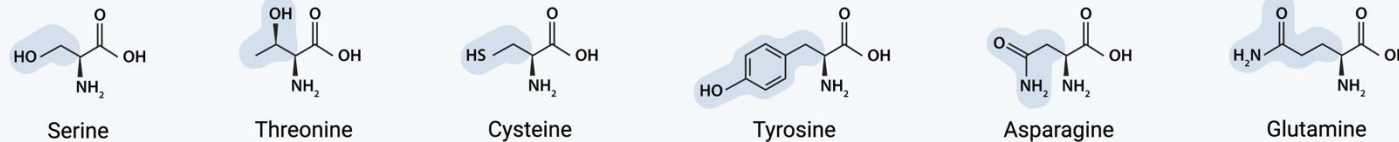
Non-polar side chains



Electrically charged side chains



Polar side chains

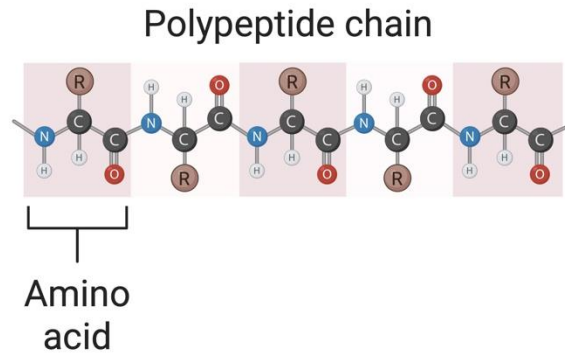


- **Non-polar side chains:** hydrophobic, tend to cluster inside the protein core, repel water
- **Polar side chains:** can form hydrogen bonds with water and other molecules
- **Electrically charged side chains:** carry a net charge at physiological pH — acidic (negative) or basic (positive)

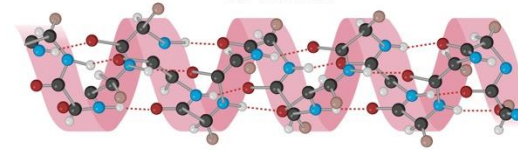
Protein structure organization

Primary

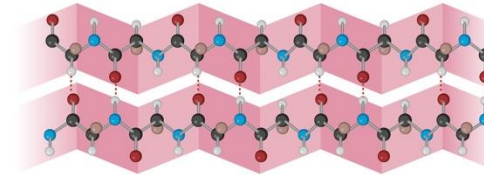
The linear sequence of amino acids joined by peptide bonds



α -helix



β -pleated sheet

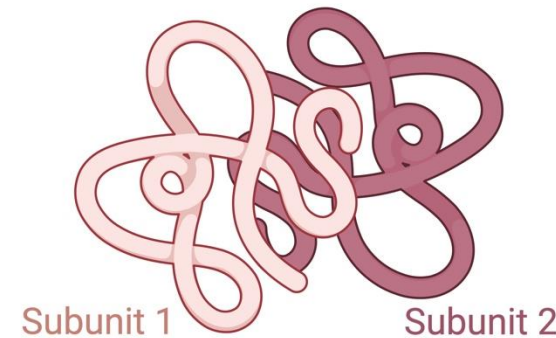
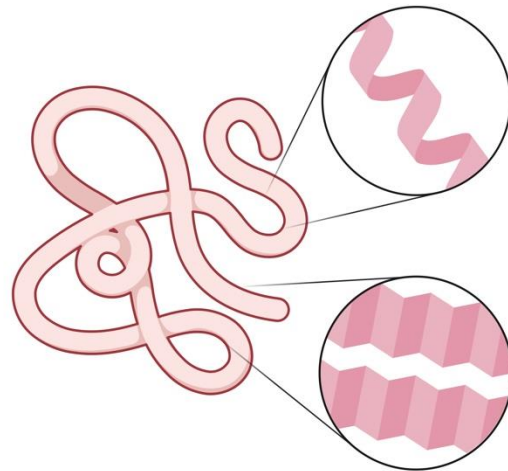


Secondary

Hydrogen bonds between backbone atoms stabilize the folded structure of the polypeptide chain

Tertiary

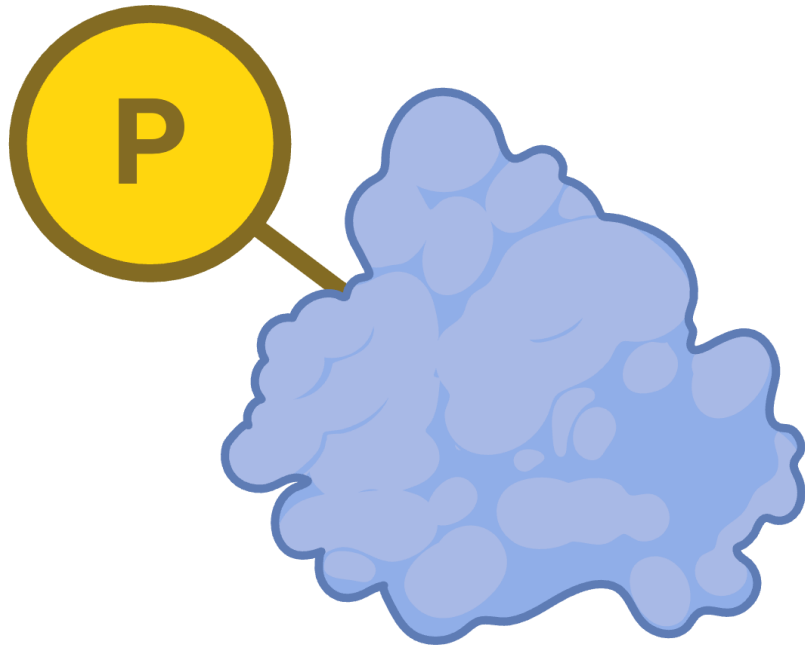
The entire chain folds on itself into a specific 3D shape



Quaternary

Some proteins work only as an assembly of different subunits

Beyond sequence



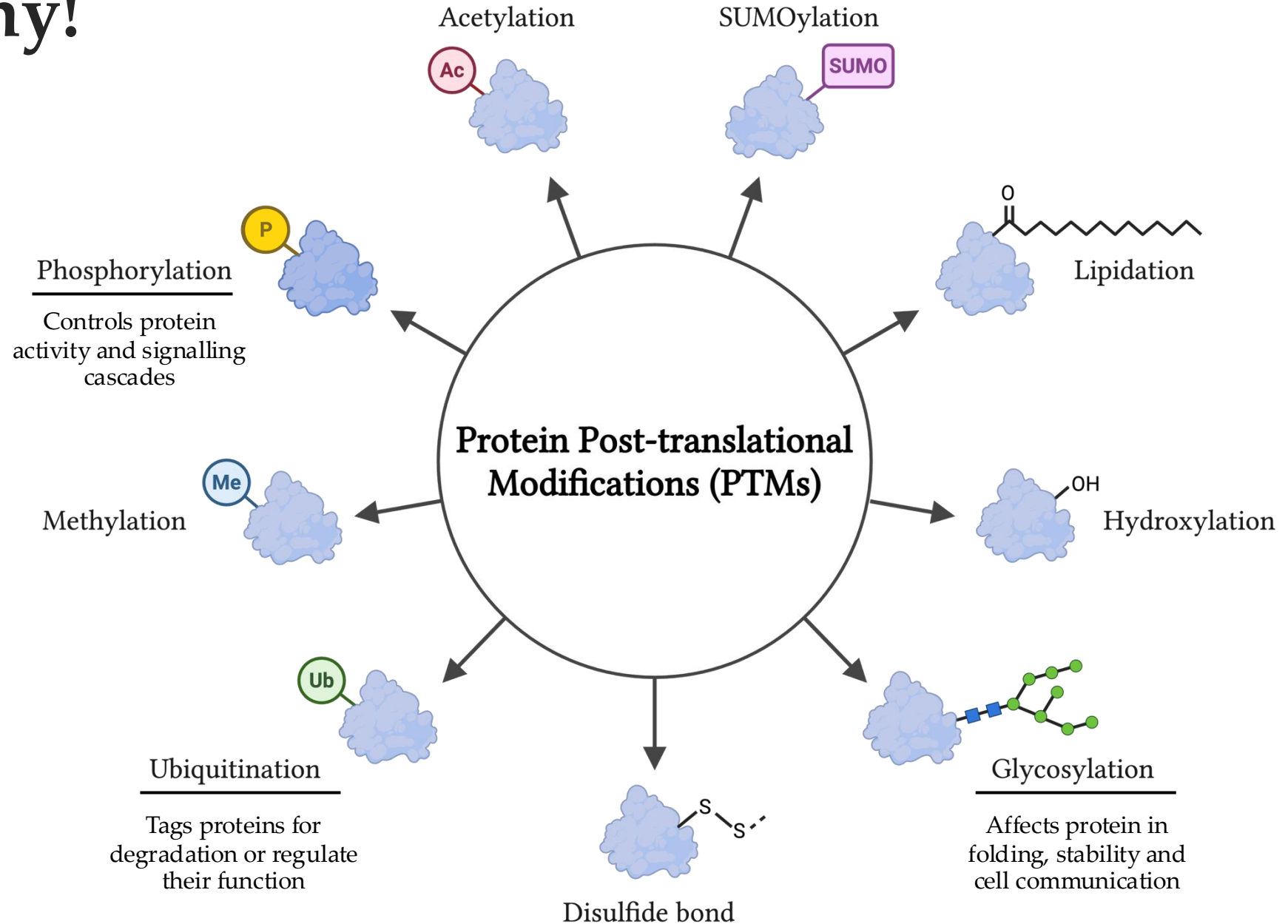
Post-translational modifications (PTMs) are **chemical groups** added to amino acid side chains after translation, altering the protein's charge, shape or interactions.

PTMs increase the **complexity** of the proteome.

A single gene can give rise to dozens of distinct protein forms depending on which modifications are present.

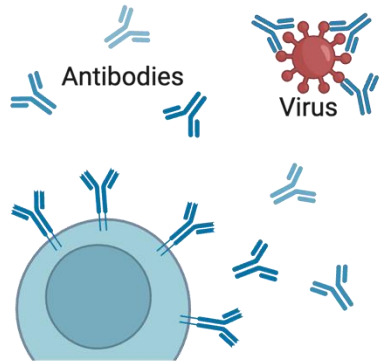
Oh no! how many!

These three PTMs are particularly common in proteomics studies because they are **abundant**, **functionally critical** and **detectable** in proteomics experiments

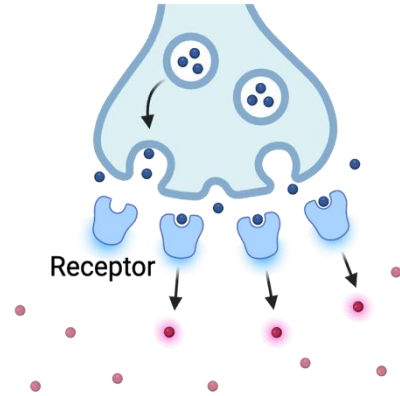


Proteins are like PhD students: very busy...

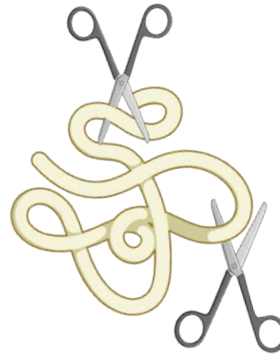
Defensive proteins



Receptor proteins



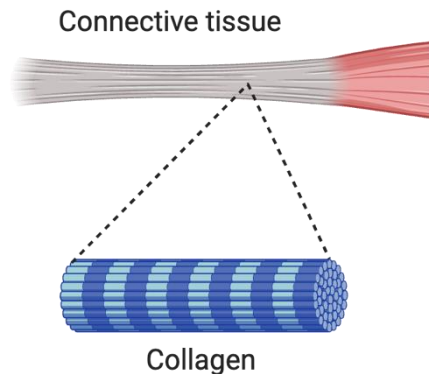
Enzymatic proteins



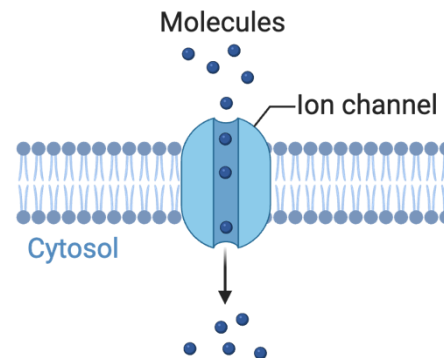
Examples

- **Defensive proteins:** antibodies (IgG) neutralize pathogens and foreign molecules
- **Receptor proteins:** insulin receptor binds insulin and triggers glucose uptake signaling
- **Enzymatic proteins:** trypsin cleaves peptide bonds; DNA polymerase replicates DNA
- **Structural proteins:** collagen provides tensile strength in connective tissue; actin forms the cytoskeleton
- **Transport proteins:** hemoglobin carries oxygen in red blood cells; albumin transports fatty acids in plasma

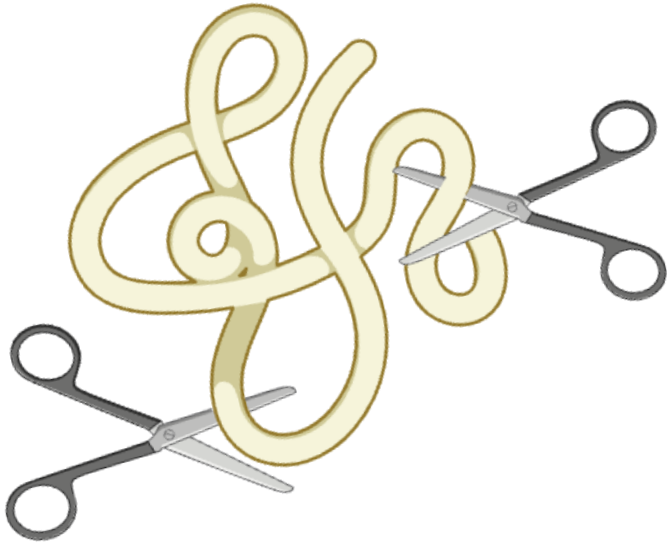
Structural proteins



Transport proteins



From proteins to peptides...

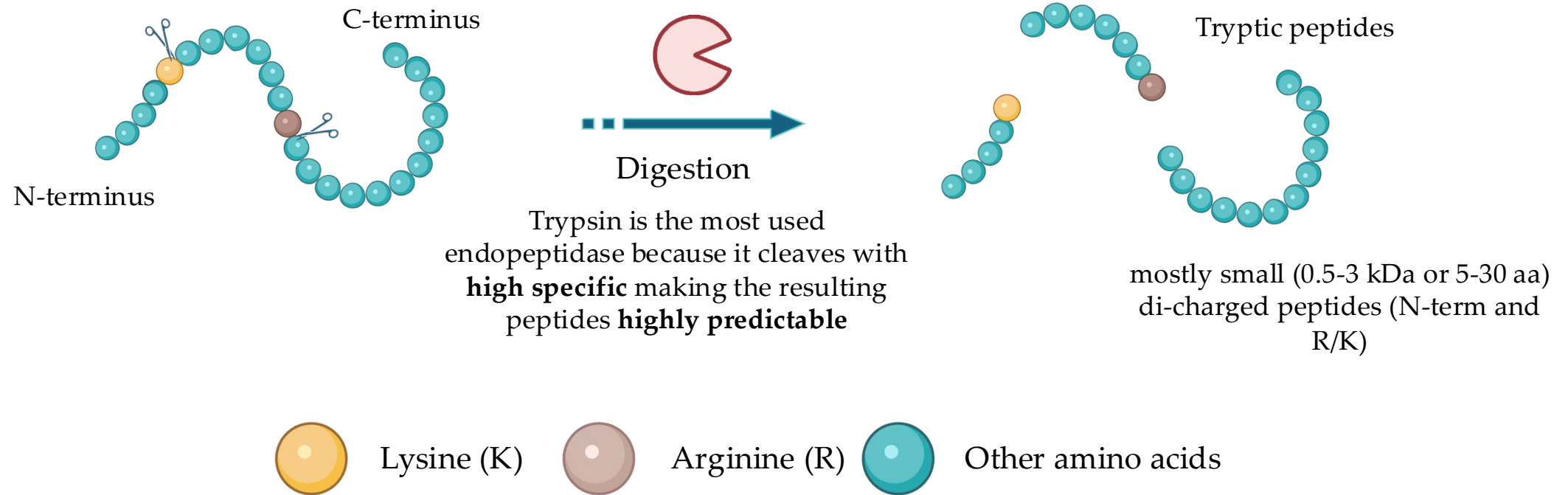


Proteases are enzymes that cleave proteins into smaller peptides by cutting specific peptide bonds.

Each protease has its own **cleavage specificity**, depending on the amino acid sequence!

Trypsin: the proteomics gold standard

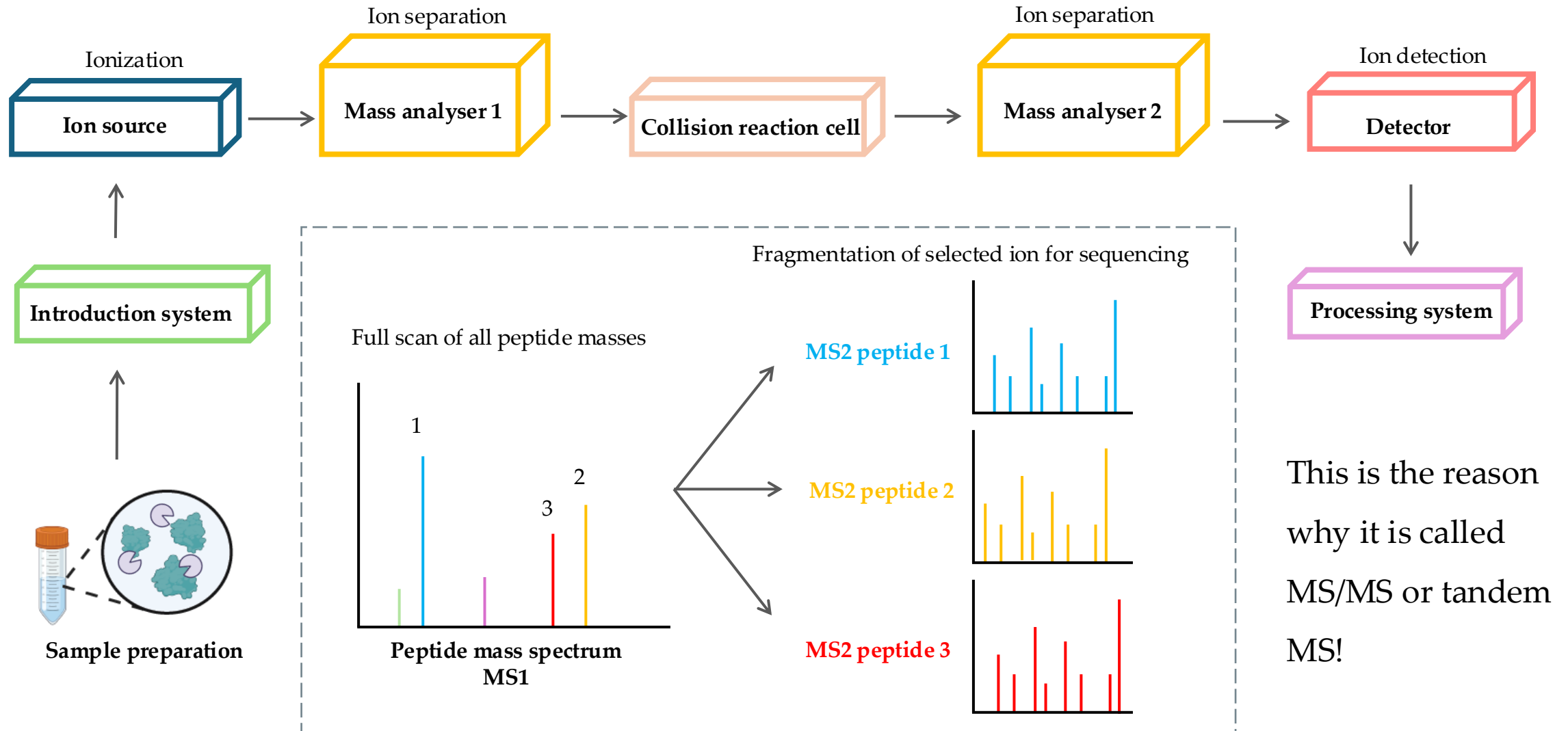
Trypsin cleaves after lysine (K) and arginine (R), unless there is a proline (P)



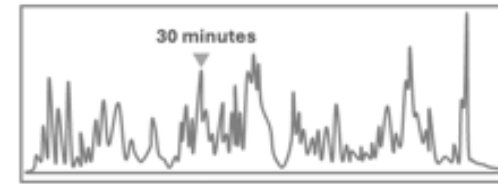
Missed cleavages occur when trypsin fails to cut at a K or R site.

This can happen due to adjacent proline residues, nearby PTMs, local protein structure, or simply incomplete digestion.

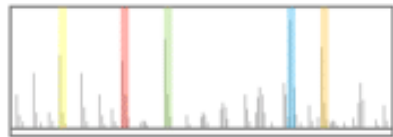
How mass spectrometry works?



DDA and DIA acquisition methods



Data Dependent Acquisition (DDA)
selects ions based on intensity



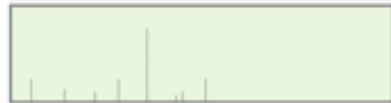
Precursor scan (MS1)
Fragments only a subset of peptides



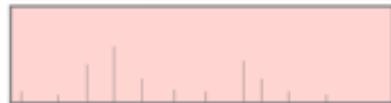
MS/MS orange peptides



MS/MS blue peptides



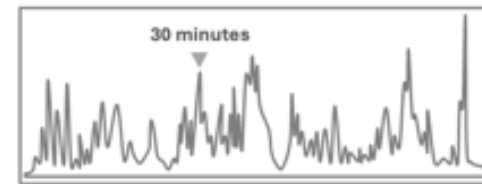
MS/MS green peptides



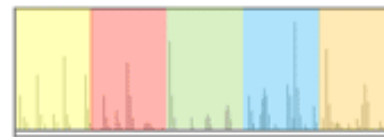
MS/MS red peptides



MS/MS yellow peptides



Data Independent Acquisition (DIA)
fragments all ions in a defined m/z
window.



Precursor scan (MS1)
Fragments all peptides



MS/MS of peptides in orange window



MS/MS of peptides in orange window



MS/MS of peptides in orange window

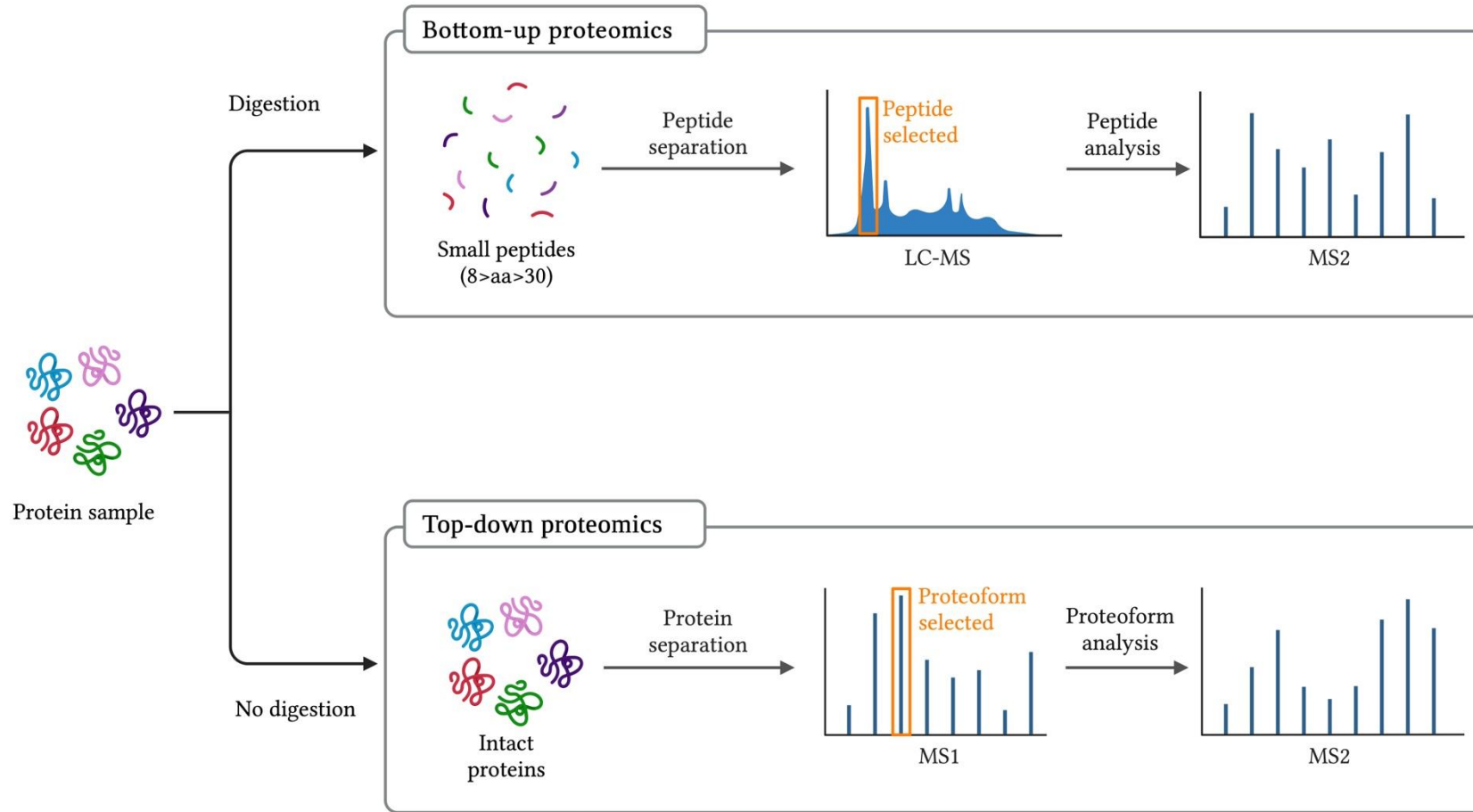


MS/MS of peptides in orange window



MS/MS of peptides in orange window

Bottom-up proteomics



Bottom-up proteomics is currently the dominant approach, as it is more compatible with existing MS instruments and requires simpler sample preparation compared to **top-down** proteomics

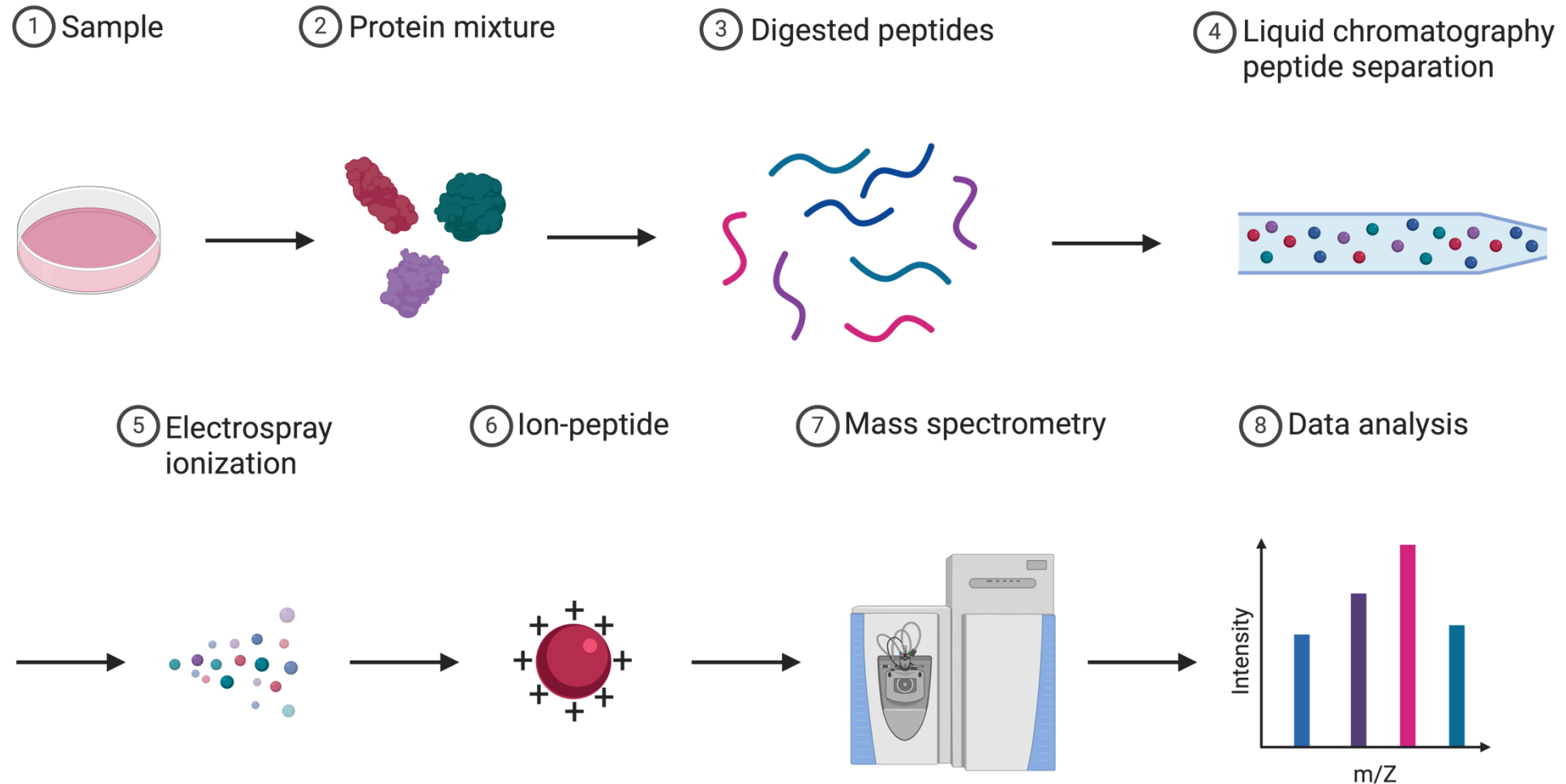


Break!

Can you trace the journey of a protein
from sample to detected mass?

Quick recap!

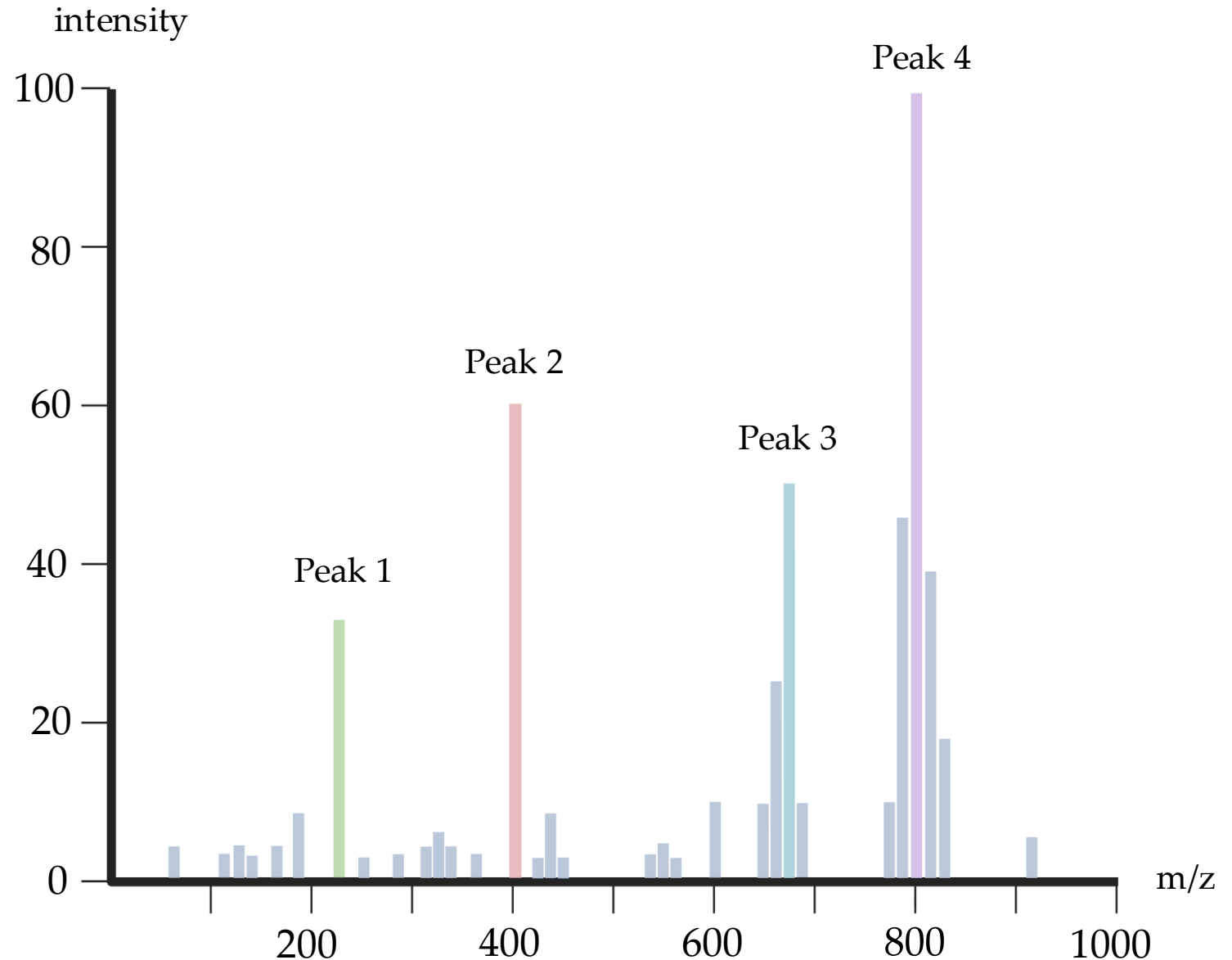
Peptides are separated by **hydrophobicity**: the more hydrophobic a peptide, the longer it takes to elute from the column.



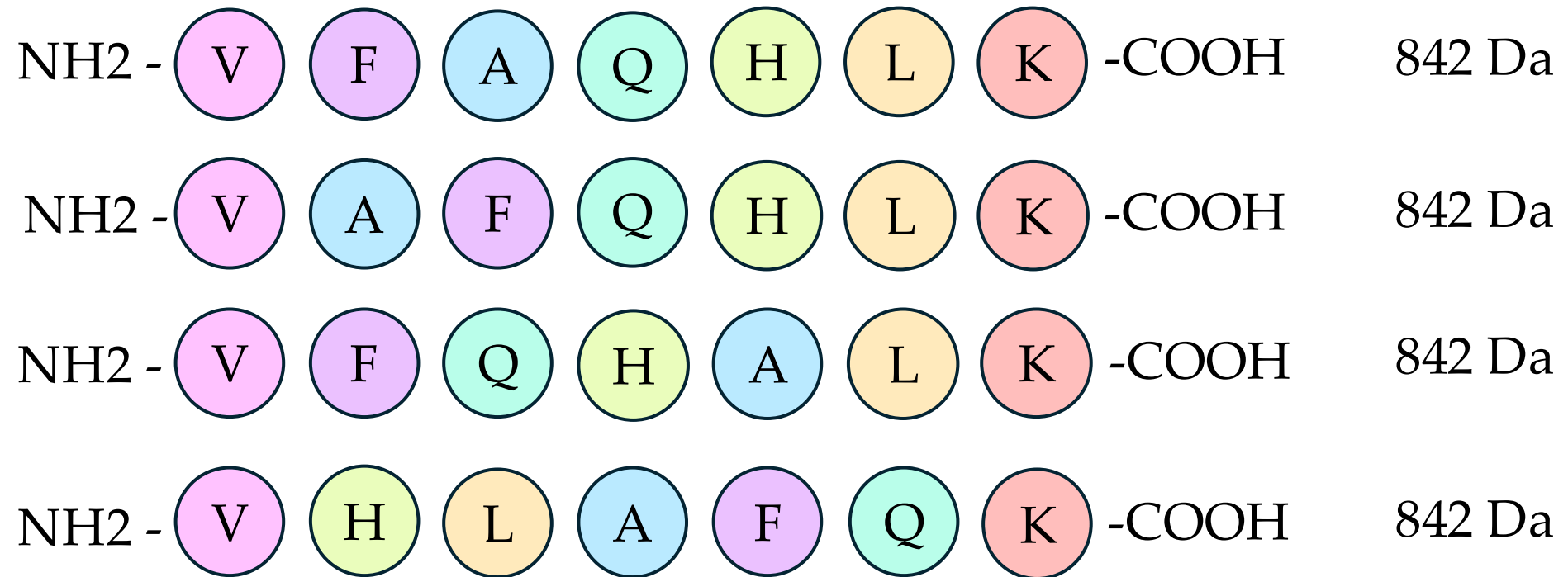
Let's look at MS1

- How abundant that specific ion was in the sample at the time of detection
- Mass-to-charge ratio (m/z) is calculated by dividing the ion's mass by its charge. Each peak position along the axis tells you the size of the detected ion

Example: a peptide of 1000 Da with charge 2+ is detected at m/z 500.

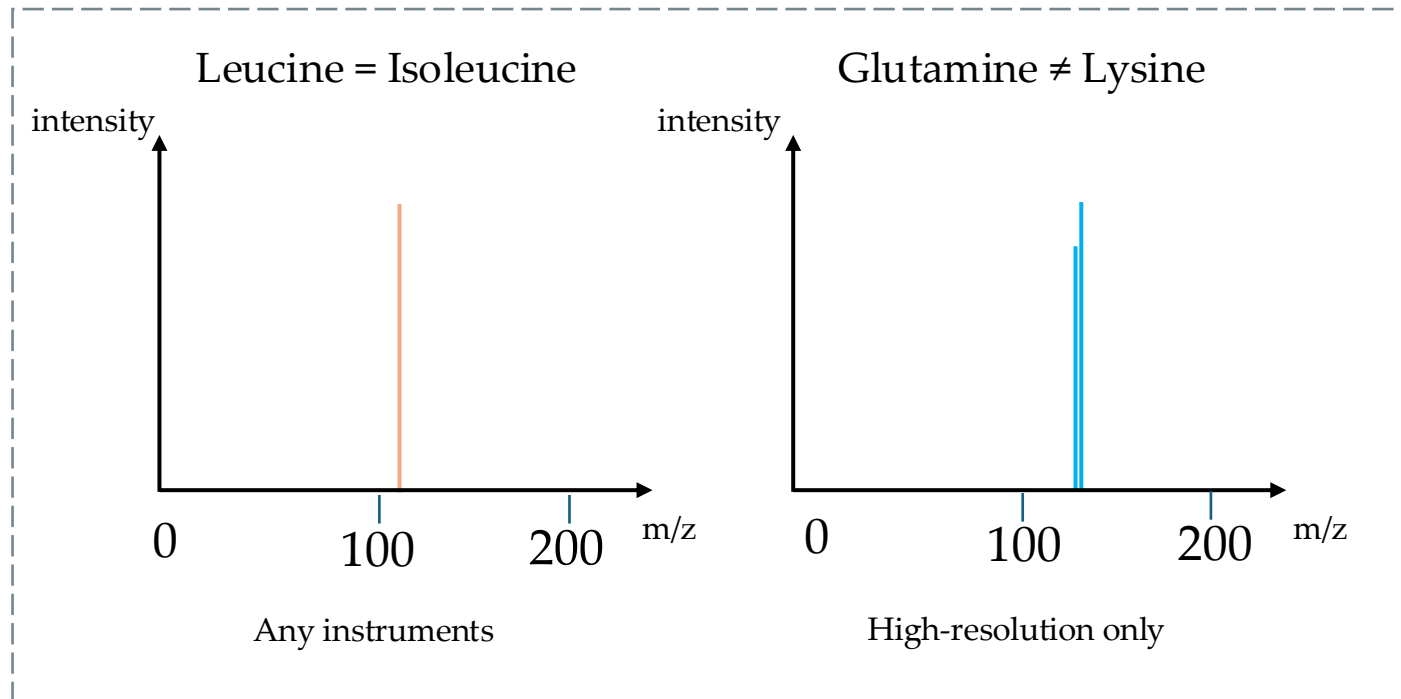


Same masses, different sequences...



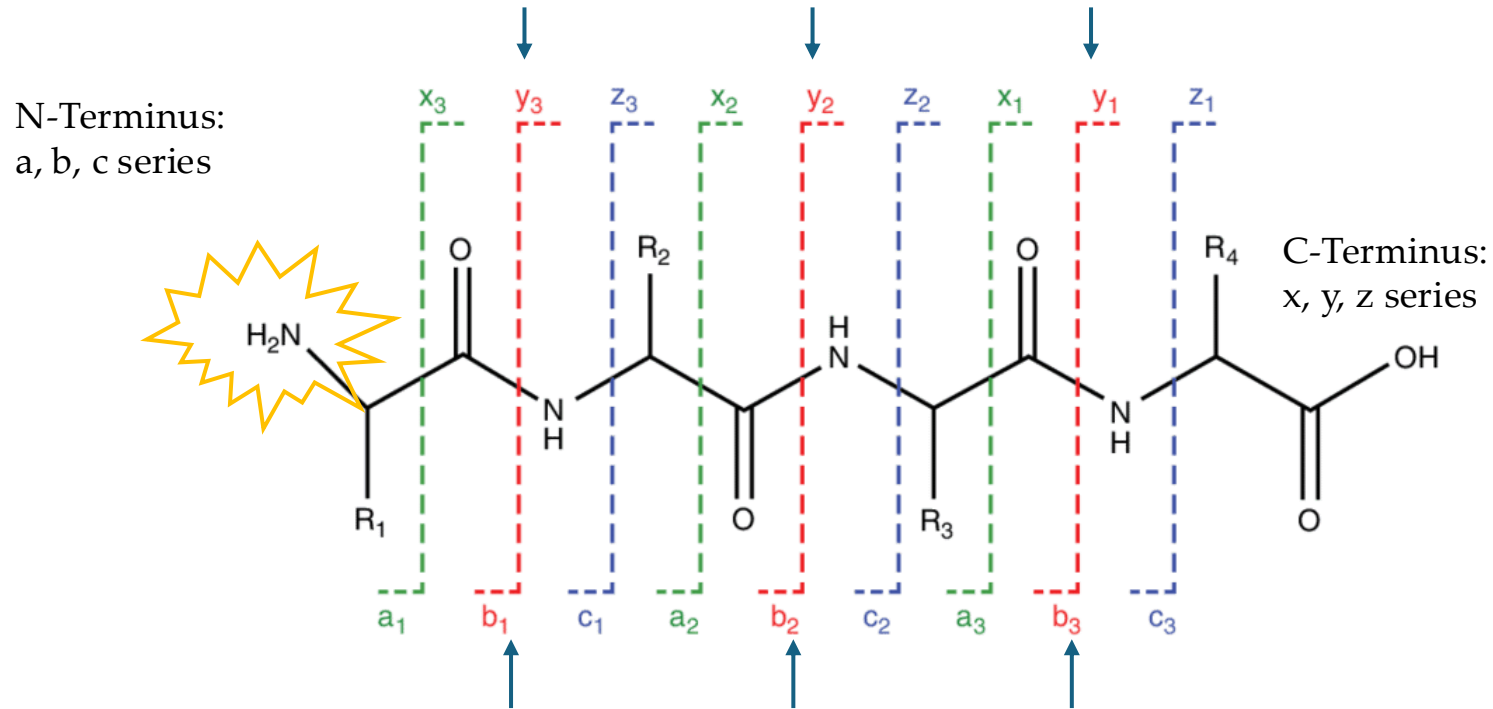
Mass accuracy and instrument resolution matter

- Leucine and Isoleucine share an identical residue mass and cannot be distinguished by standard MS/MS
- Glutamine and Lysine are nearly isobaric (128.13 vs 128.17 Da):
only high-resolution instruments like Orbitrap can resolve them



Amino Acid	Abbreviation	Residue Mass (Da)
Glycine	G	57.05
Alanine	A	71.08
Serine	S	87.08
Valine	V	99.13
Leucine	L	113.16
Isoleucine	I	113.16
Threonine	T	101.11
Cysteine	C	103.15
Proline	P	97.12
Phenylalanine	F	147.18
Tyrosine	Y	163.18
Tryptophan	W	186.21
Aspartic Acid	D	115.09
Glutamic Acid	E	129.12
Asparagine	N	114.10
Glutamine	Q	128.13
Lysine	K	128.17
Arginine	R	156.19
Histidine	H	137.14
Methionine	M	131.19

Peptide fragmentation patterns



Only fragments that carry a charge can be revealed by the detector!

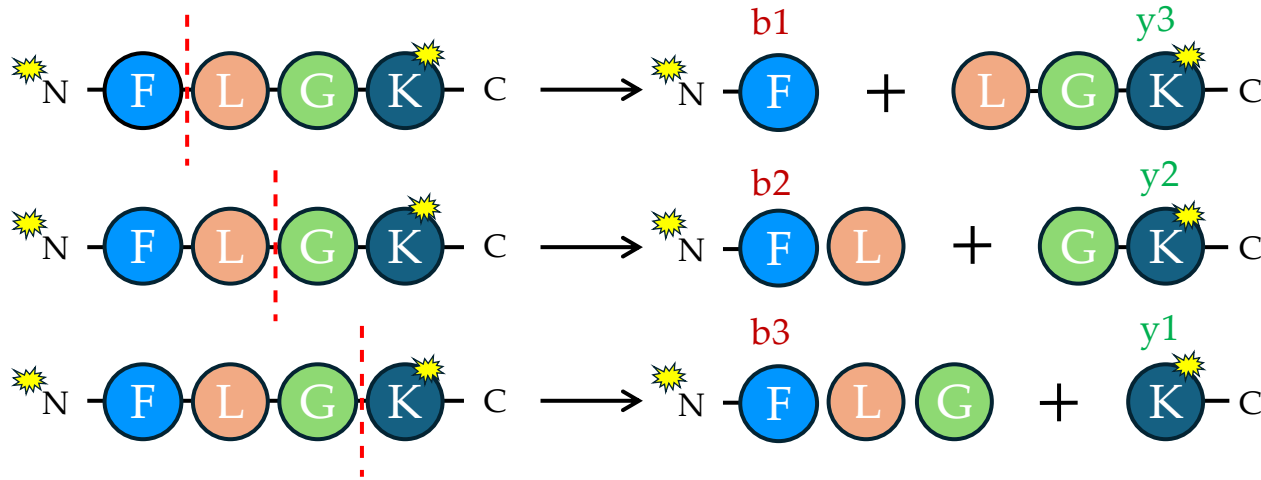
Break point around the peptide bond depends on the **fragmentation method**.

The most common fragmentation methods (CID/HCD) produce mostly **b/y ions**.

Peptide fragmentation can be controlled by instrument parameters (**pressure, collision energy ...**)

Fragment ions reveal the peptide sequences

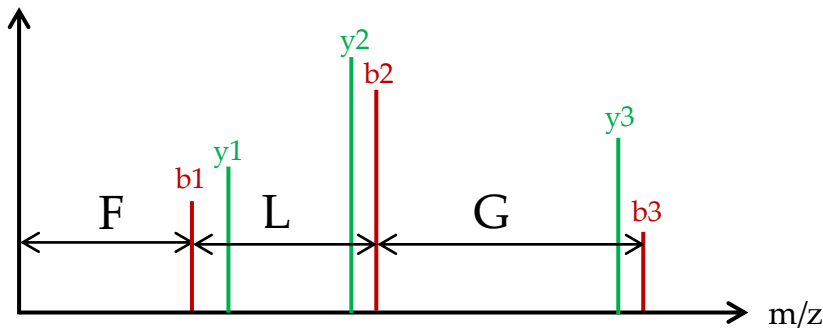
Fragmentation site



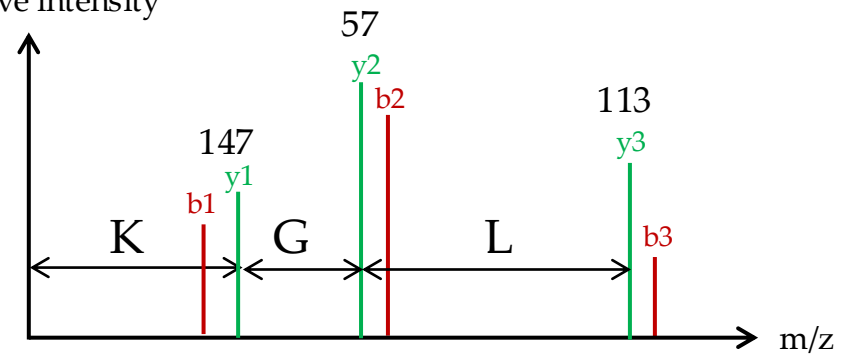
Example

1. Practical to start with y1: Lysine (147) or Arginine (175)
2. y1 may not be observed
3. b1 almost never observed
4. Leu/Ile are isobaric
5. Read the sequence from the end

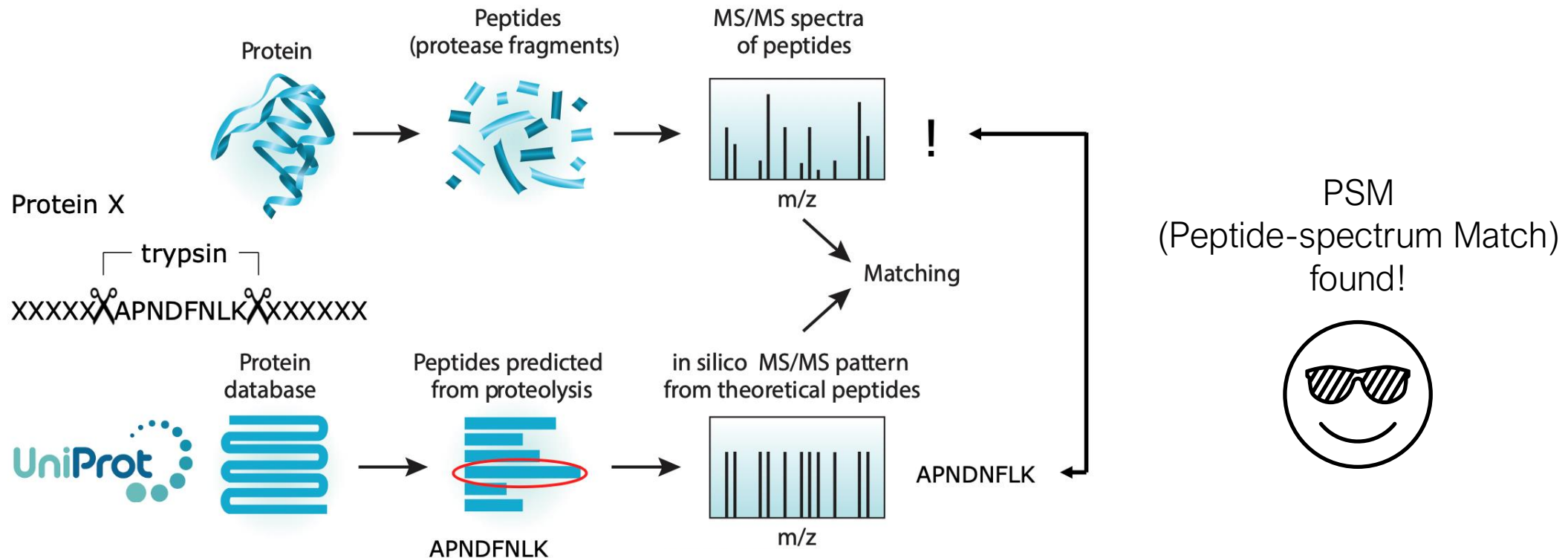
Relative intensity



Relative intensity

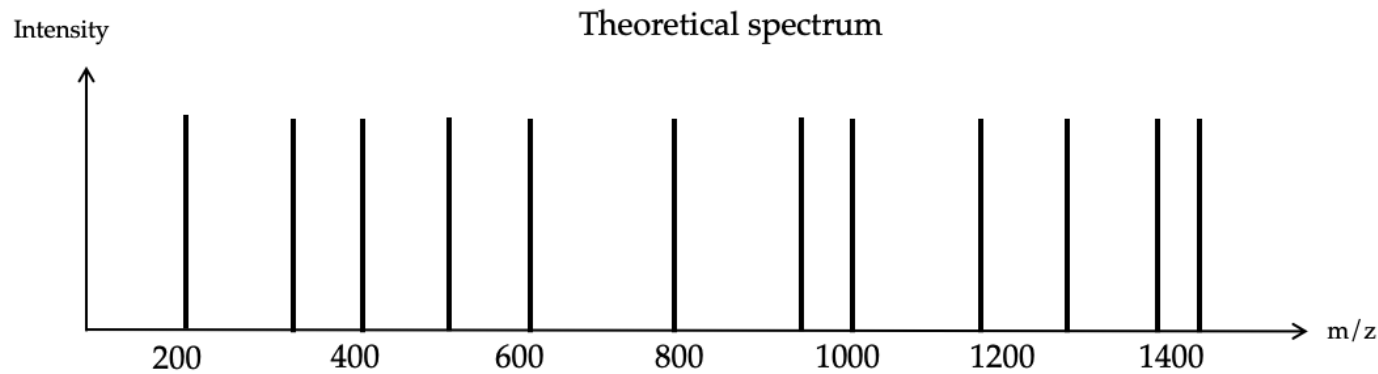


Database search helps to identify peptides

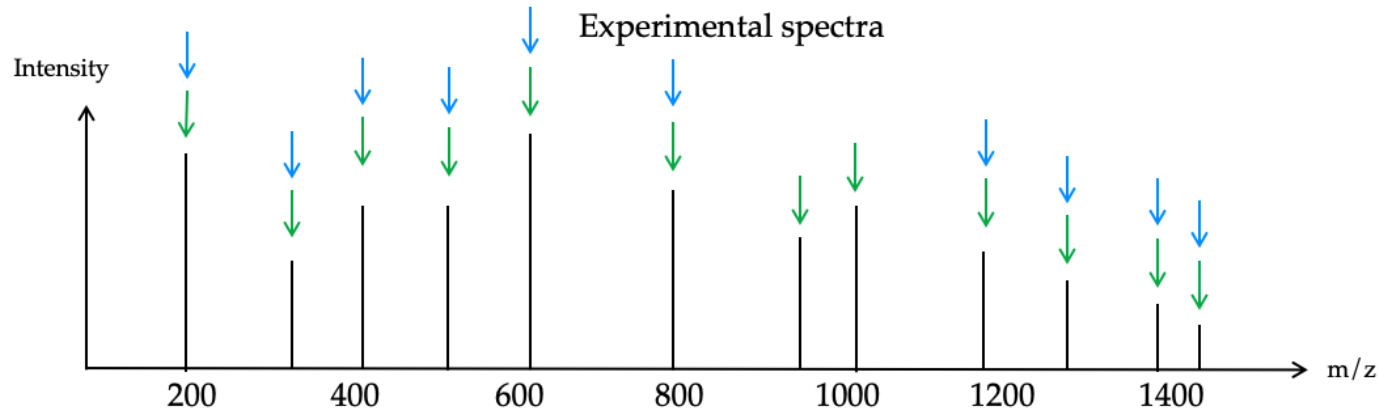


Unfortunately, we will always produce matches in large scale analysis, even if they might be wrong!

Mascot ion score (local check)



The experimental spectrum is matched against the theoretical spectrum predicted from the database.



The more peaks align, the higher the score!

Ranking PSM: how do we choose the best match

PEP 1	#	Y ions
T	13	
A	12	1461.8
D	11	1390.8
K	10	1275.7
N	9	1147.6
V	8	-
T	7	-
F	6	777.46
L	5	630.39
Q	4	517.3
T	3	389.3
L	2	288.2
R	1	175.1

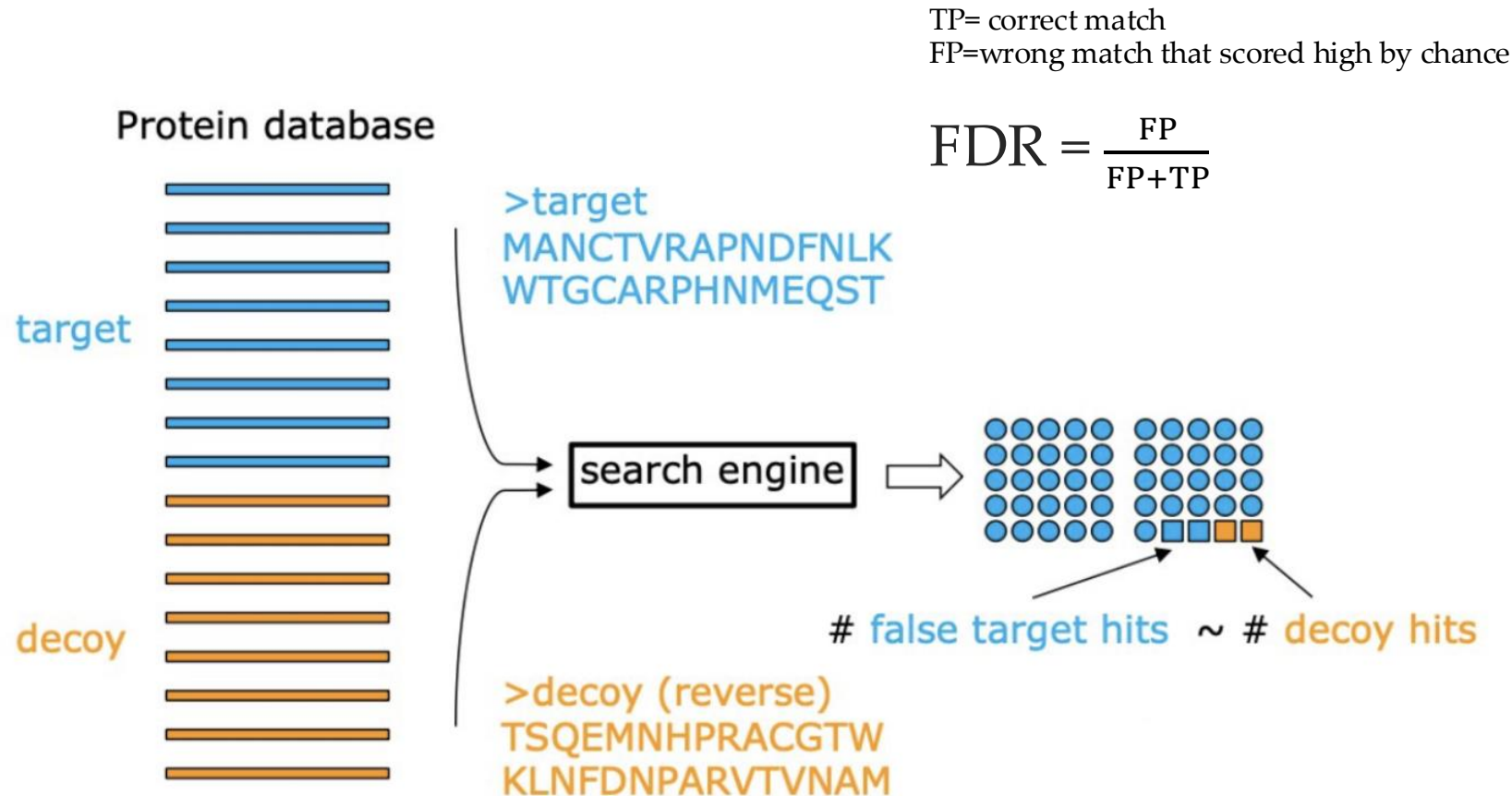
PEP 2	#	Y ions
T	13	
A	12	1461.8
D	11	1390.8
K	10	1275.7
N	9	1147.6
Q	8	1033.6
K	7	905.5
F	6	777.46
L	5	630.39
Q	4	517.3
T	3	389.3
L	2	288.2
R	1	175.1

Rank	Peptide	Mascot Ion Score
1	TADKLQKFLQTLR	225
2	TADKNVTFLQTLR	152
3	TANELQEFLQTLR	89
4	...	...

PEP2 has more matching fragment ions,
making the match statistically more significant

Target-decoy and FDR

Usually, we deal with more than 100'000 PSMs, not all of them are good quality, resulting in a distribution of high and low scoring PSMs.



How do we decide which peptides are wrong or right?

FDR permits to **control the number of incorrect matches** and so minimise mistakes!

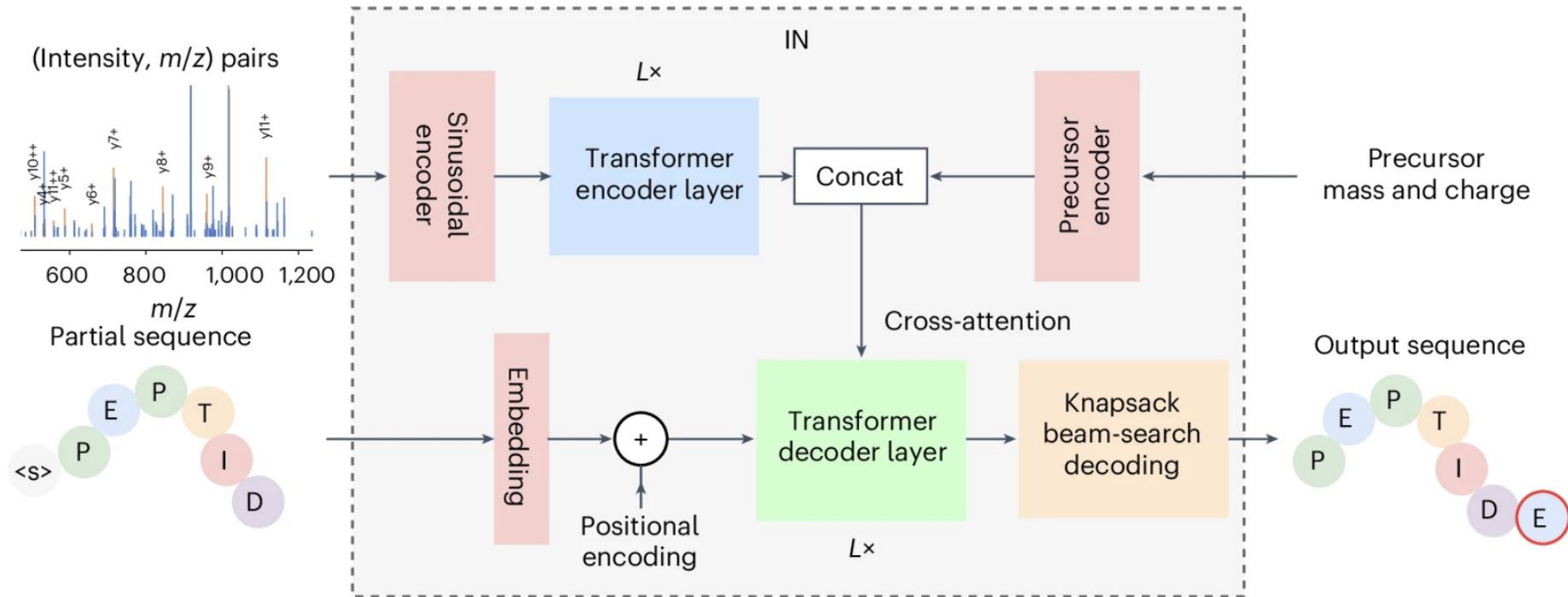
Example

If we have 1000 PSMs and 30 decoy hits, $\text{FDR} = 30/1000 = 0.03$

1% FDR is the standard

Reference-free peptide sequencing

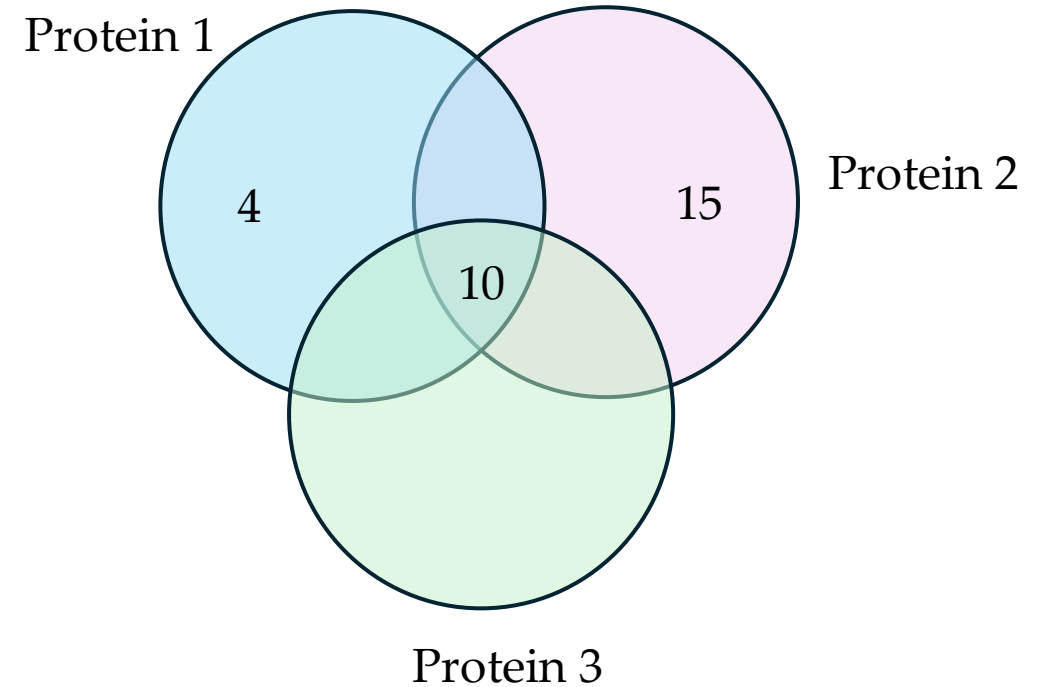
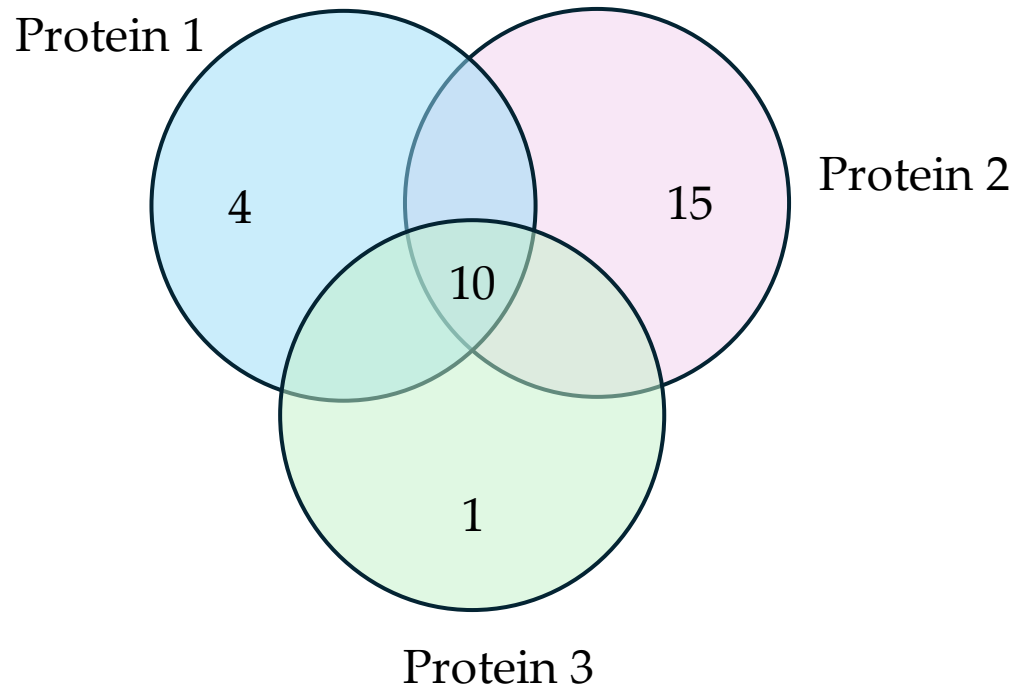
We can identify peptide sequences directly from MS2 spectra, without matching against any protein database!



The protein inference problem

Many proteins share identical sequence parts, so proteins can only be identified if a unique peptide is present!

In **quantification** shared peptides complicate the assignment of intensities to specific proteins



No info about presence or absence of protein 3

Roadmap



- Omics sciences
- Proteomics
- Amino acids
- Protein structure
- Post-translational modifications
- Protein functions

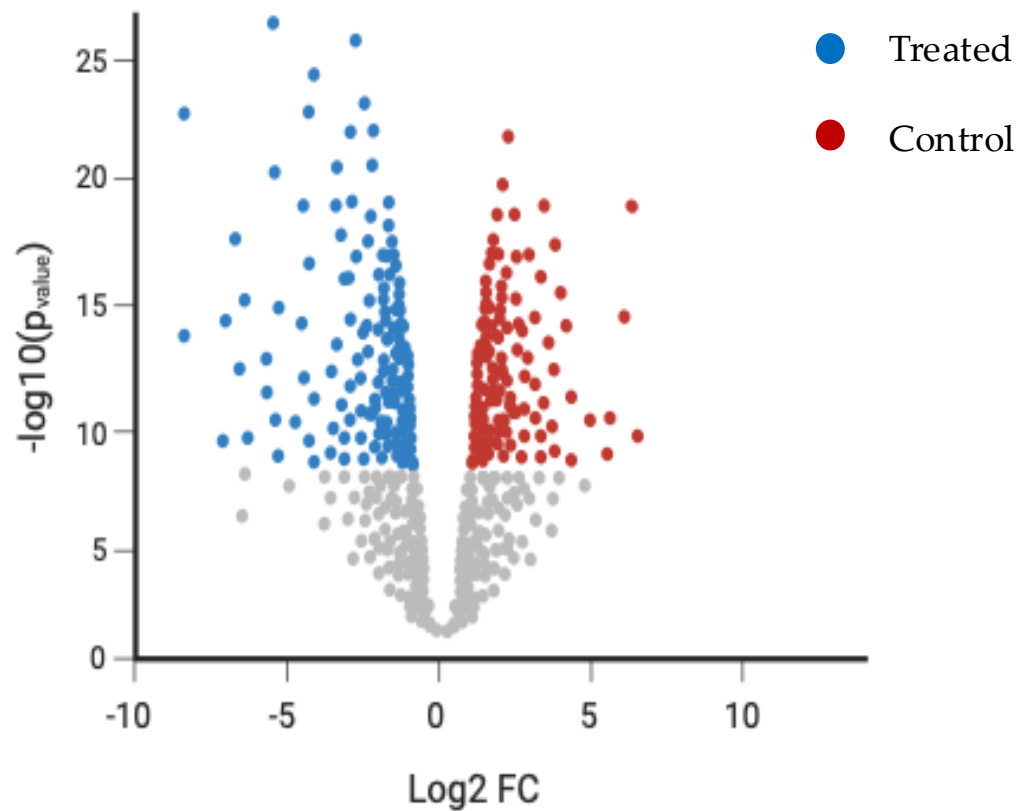
- Trypsin
- From protein to peptides
- Mass spectrometry
- Bottom-up proteomics
- DDA and DIA acquisition methods

- Precursor mass
- From MS1 to MS2
- Fragmentation patterns
- Manual sequencing
- Database search
- De novo sequencing

- Relative/absolute quant.
- Third dimension: time
- LFQ
- Normalization
- TMT
- What's next

Protein quantification

Protein quantification is the measurement of the abundance of proteins in a sample, allowing us to detect which proteins change and by how much between different biological conditions



The **x-axis** shows how much a protein increases or decreases between conditions

A value of 1 means the protein is 2× more abundant; -1 means 2× less abundant

The **y-axis** shows a statistical measure of **confidence**.

The higher the point, the less likely the result is due to chance.

Looking at this plot, which proteins would you prioritize for follow-up experiments?

Relative and absolute quantification

RELATIVE QUANTIFICATION	ABSOLUTE QUANTIFICATION
Compares same protein across conditions	Determines number of molecules
No info on absolute amount	Requires internal standards
Reports fold changes	Reports molecular counts
Good for global, untargeted profiling	Ideal for targeted, hypothesis-driven analysis
Used in discovery studies	Enables stoichiometry comparisons
Hypothesis generating	Hypothesis testing

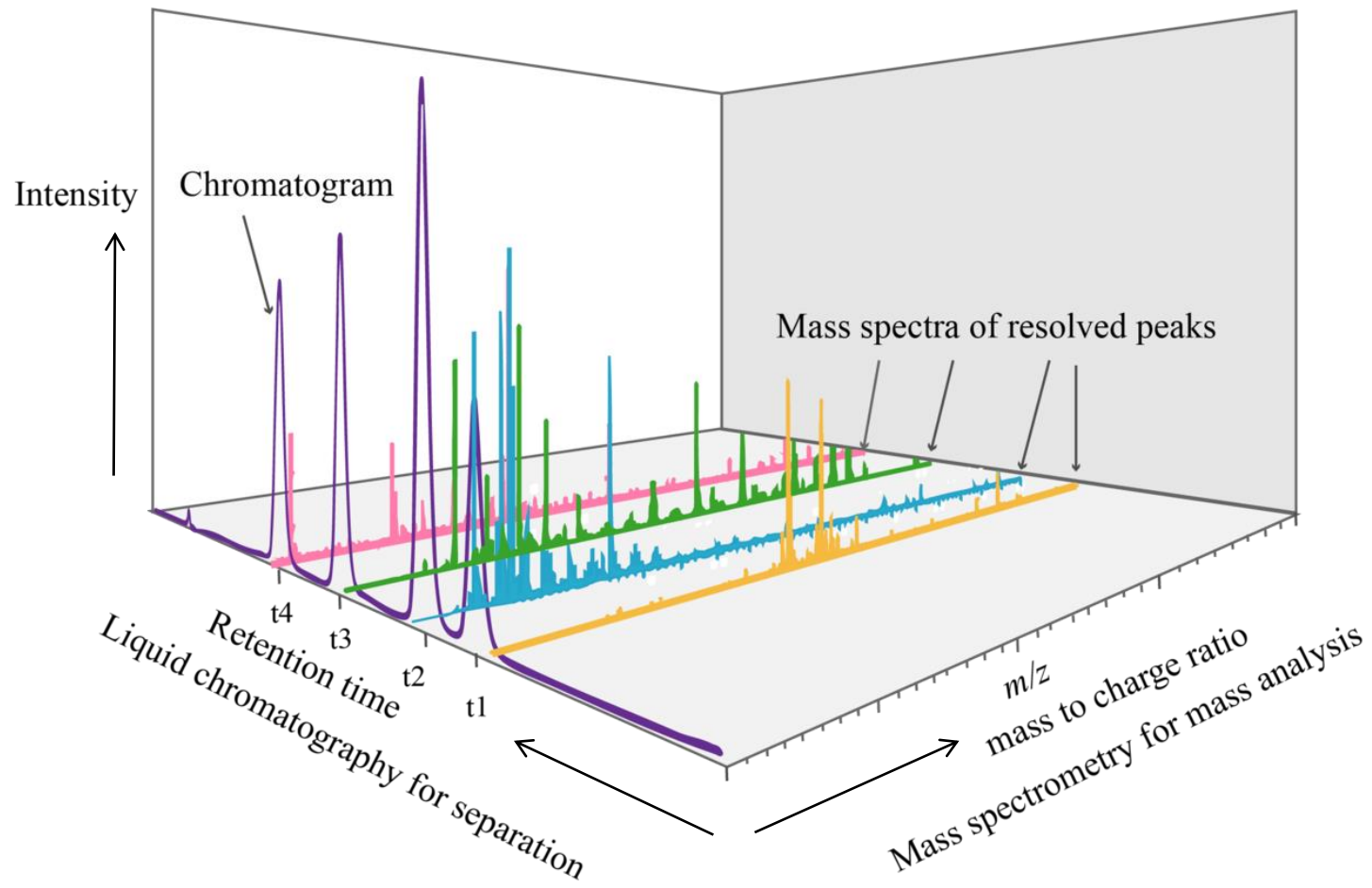
Label-free and labelled methods

Definitions

LFQ	Spectral counting
SILAC/SILAM (in vivo)	TMT/iTRAQ (in vitro)

- **Label-free:** quantification without adding chemical markers (proteins are measured directly based on their MS signal)
- **Labelled:** chemical or metabolic tags are added to peptides/proteins to distinguish samples before combining them for MS analysis
- **In vivo** (SILAC/SILAM): isotopically heavy amino acids are incorporated into proteins during cell growth (the label is added biologically, inside living cells)
- **In vitro** (TMT/iTRAQ): chemical tags are added to digested peptides after extraction (the label is added in the lab, outside the organism).

Accurate quantification requires time!



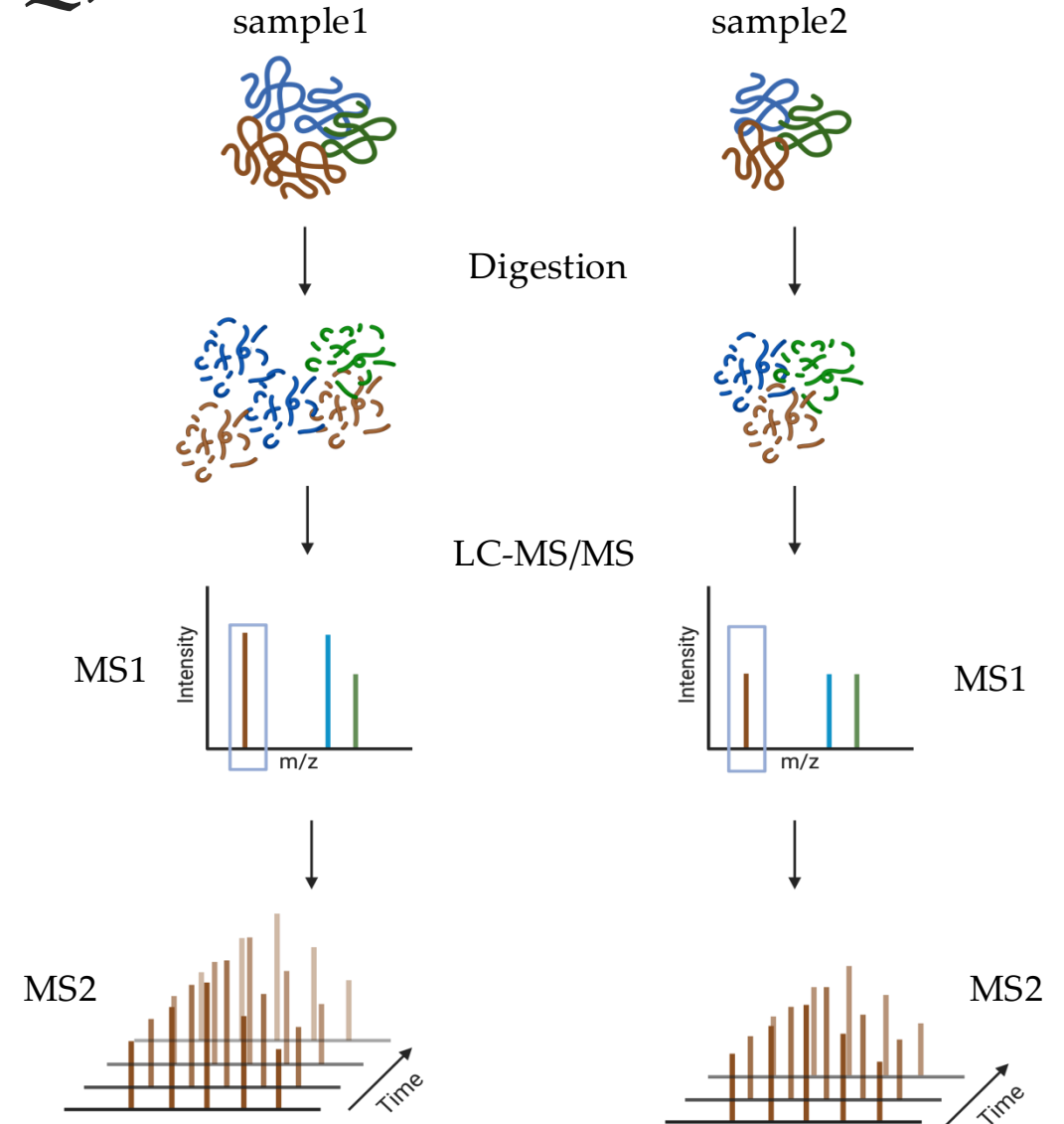
Each peptide elutes from the LC column at a characteristic time (retention time), providing a **third dimension** for quantification!

Label-free quantification (LFQ)

Because no chemical labels distinguish the samples, **intensities** must be compared across separate LC-MS runs!

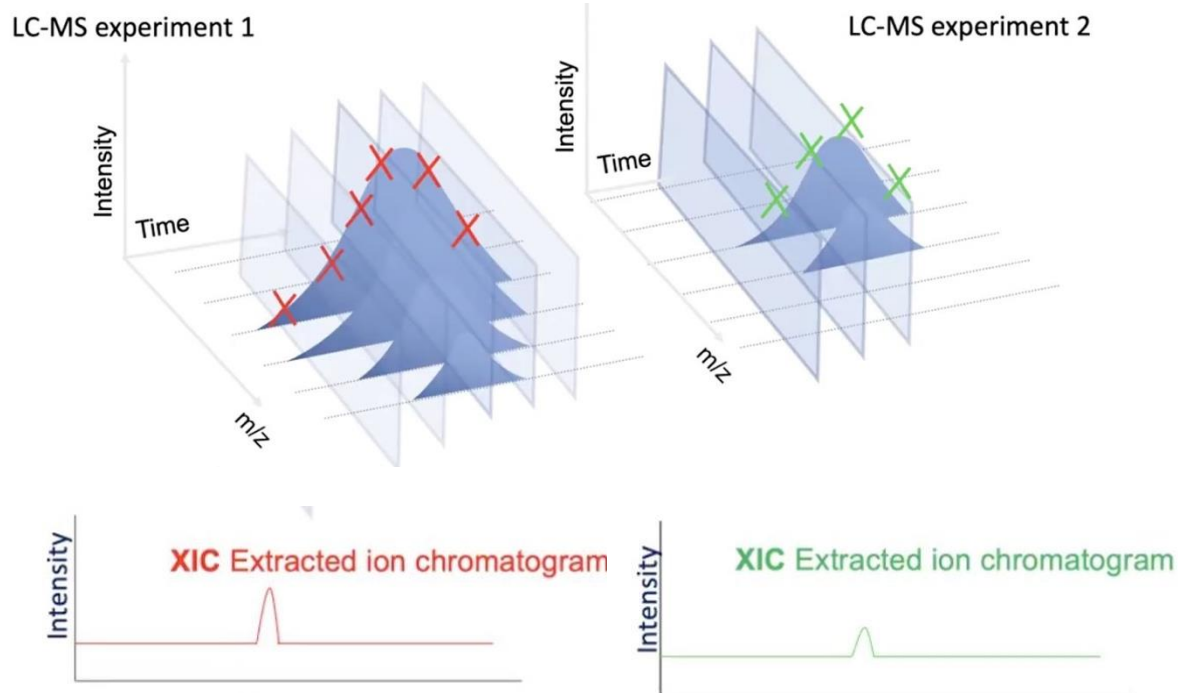
Both samples contain largely the same proteins, what changes is how much of each protein is present

- Each sample is analysed separately by LC-MS.
- Peak intensities of same ions are compared
- Identification is typically done via MS2



MaxLFQ for data normalization...

Data normalization is a **mandatory** step given that every sample is digested and injected in the LC-MS column separately, introducing a series of technical biases including **false positives**, **false negatives**, and **loss of reproducibility**



XIC = signal intensity of a specific ion over time, integrating its area gives the quantitative value.

Example

If Sample A had twice as much protein accidentally loaded, its XIC peaks will all be $\sim 2\times$ taller.

Normalization scales them down so the comparison is fair.

Pros and cons of LFQ

Cost-effective,

Simple workflow,

No chemical labelling steps

Run-to-run variability

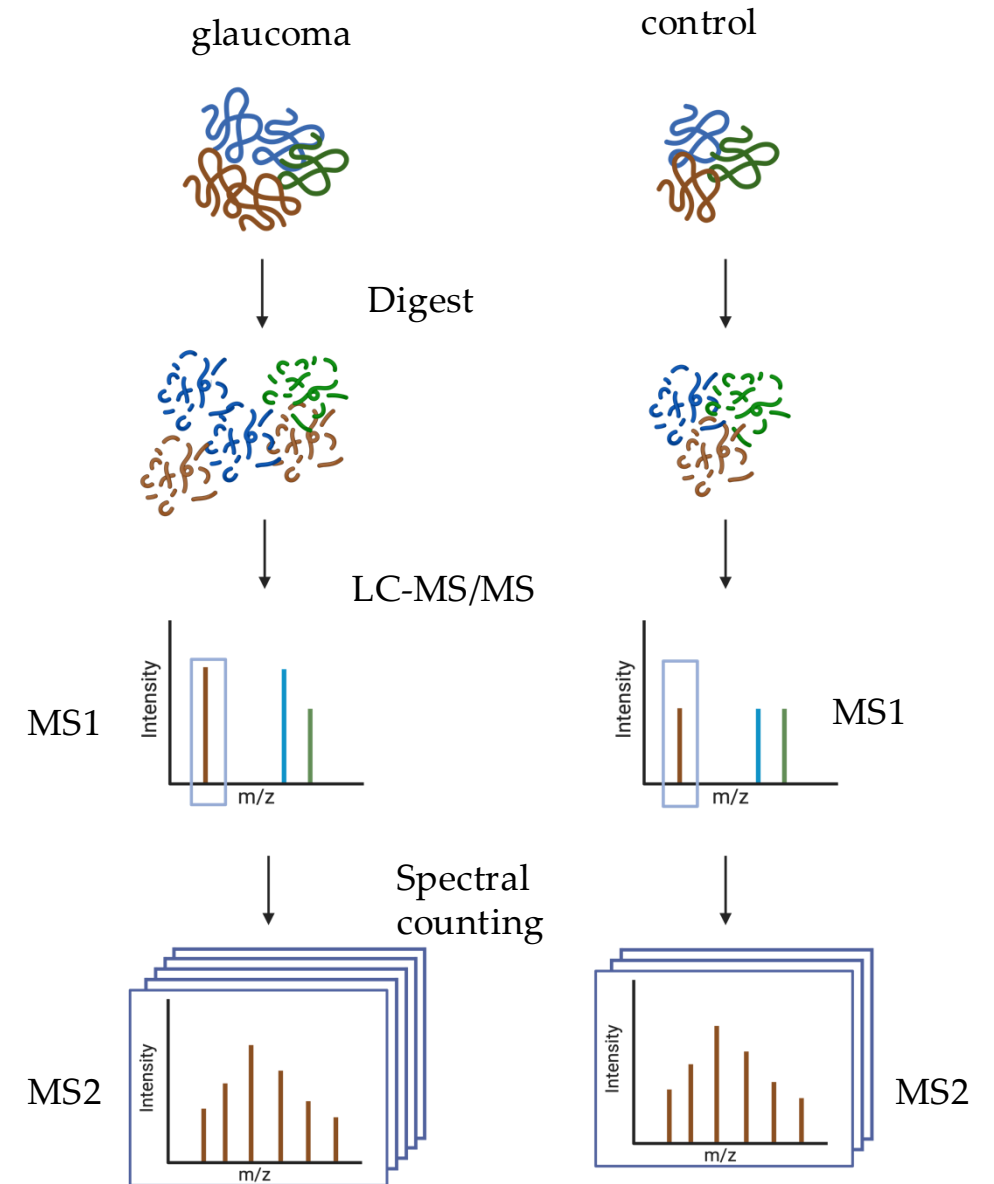
Missing values

Requires normalization

Spectral counting

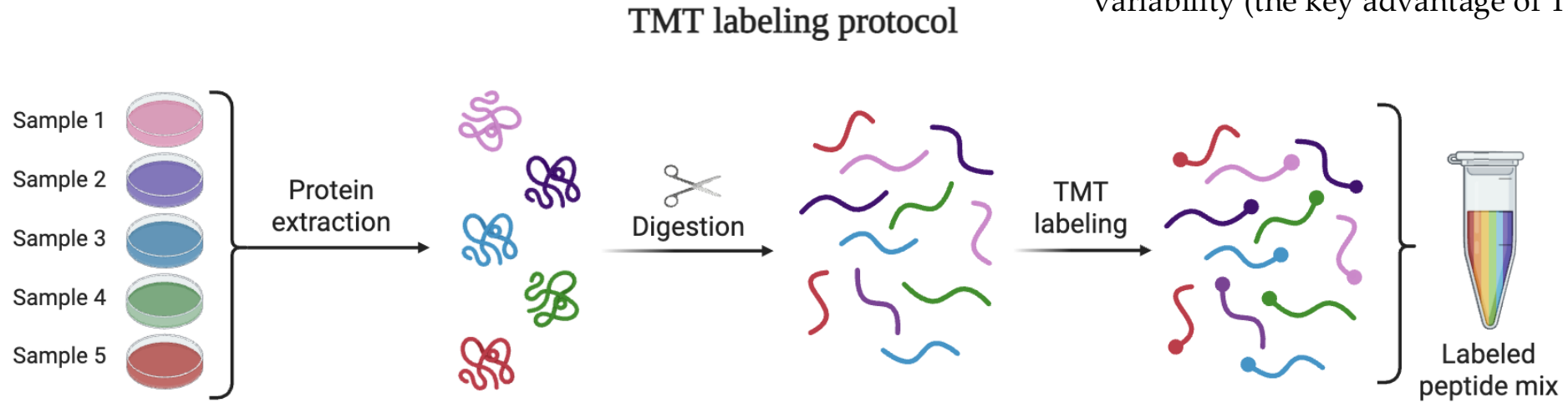
Protein abundance is estimated by counting the number of MS2 spectra (PSMs) identified for each protein. A more abundant protein will be sampled more often by the mass spectrometer, generating more PSMs.

Limitations: low accuracy for low-abundance proteins, poor dynamic range, and biased toward larger proteins (which generate more peptides and thus more spectra).

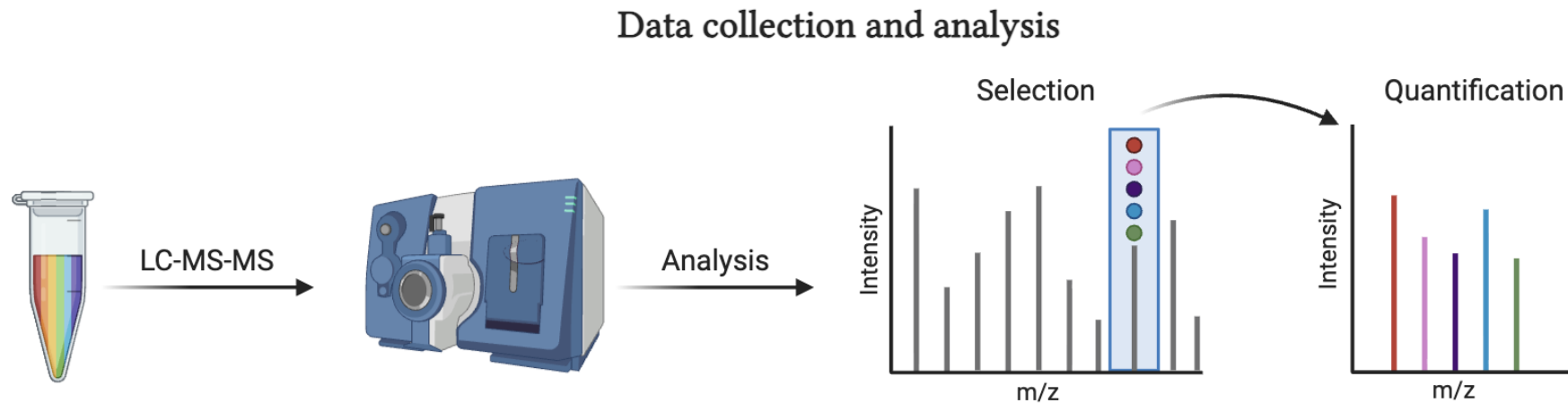


TMT labelling protocol

All samples are mixed and analyzed together in a single LC-MS run, eliminating run-to-run variability (the key advantage of TMT over LFQ)



Currently TMT reach 18-plex



End-to-end quantitative proteomics

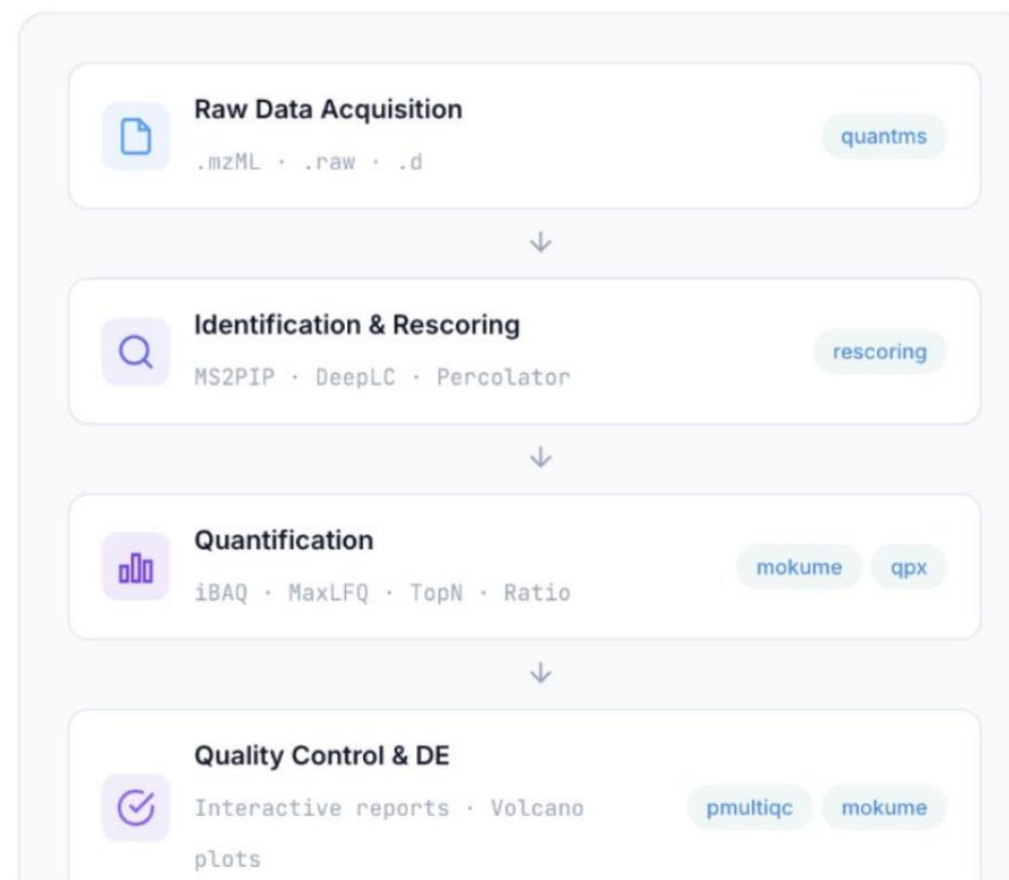
A Nextflow-orchestrated workflow where specialized tools handle each step.

These are the tools you are going to use soon!

Two dedicated Nextflow pipelines: **quantms** for DDA (LFQ & TMT/iTRAQ) and **quantmsdiann** for DIA — but the downstream tools work independently too. Use **mokume** to quantify proteins from any search engine output. Use **pmultiqc** for QC on any experiment.

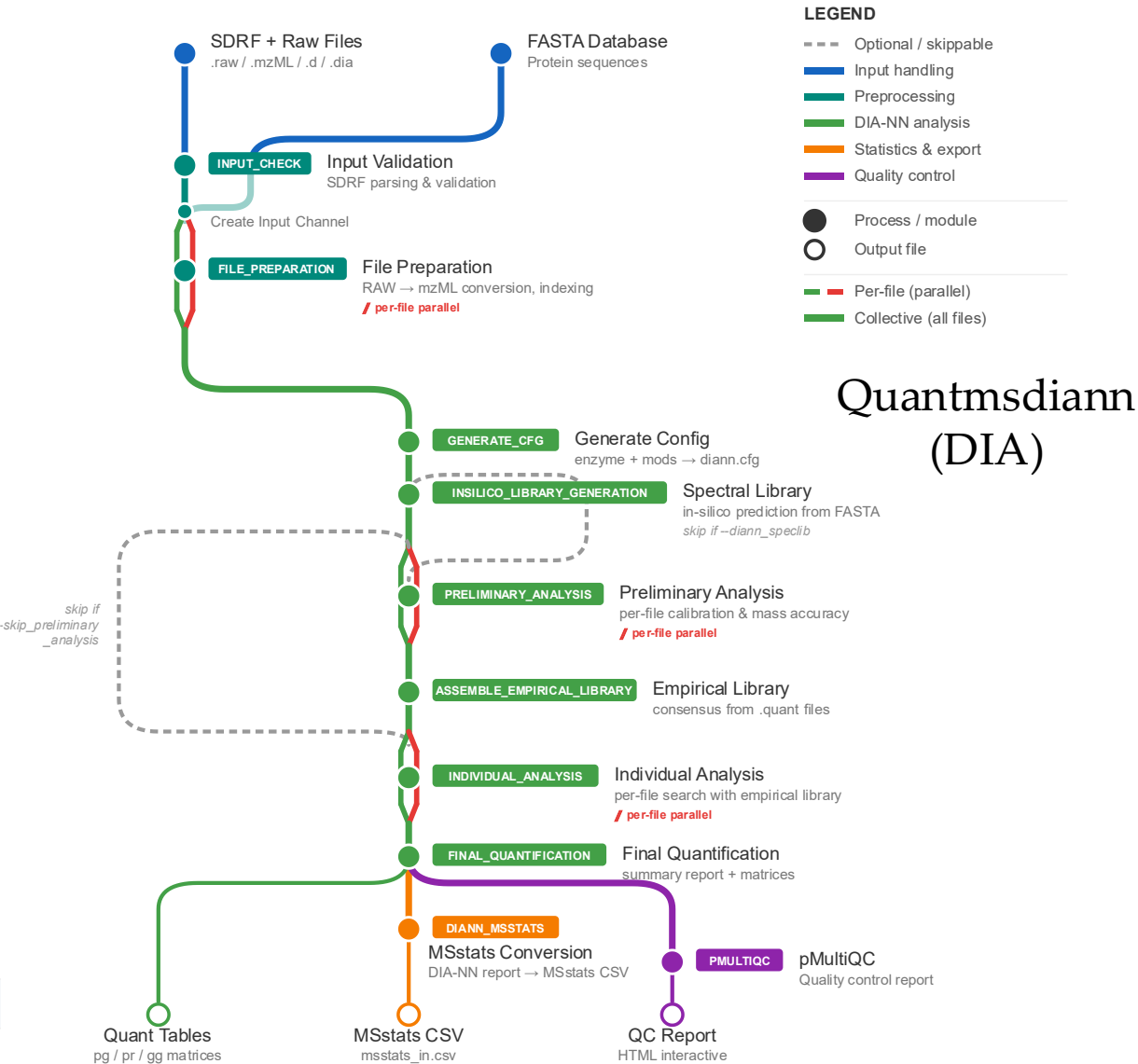
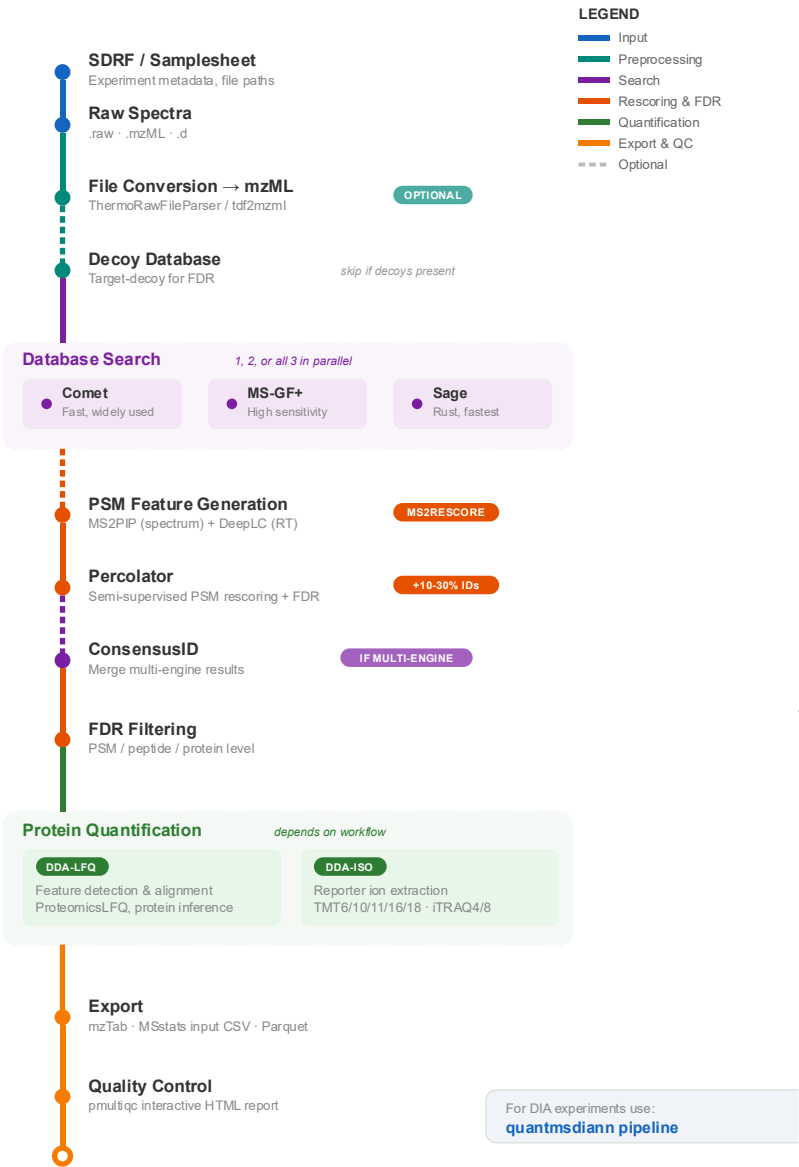
- 1 **Convert & search** raw spectra against protein databases
- 2 **Rescore PSMs** with ML models for higher sensitivity
- 3 **Quantify proteins** using iBAQ, MaxLFQ, TopN, or ratio methods
- 4 **Generate QC reports** and differential expression results

[Read pipeline docs →](#)



Soon...

QuantMS
(DDA)



Before the break please create your codespace!

You can find the instruction at the following link:

https://biosustain.github.io/dsp_course_proteomics_intro/material/instructions_quantms_PXD04621.html#run-the-analysis

Mass spectrometry-based
Proteomics introduction
documentation



Search

% + K

QuantMS

QuantMS Hands-On for
PXD04621

QuantMS Input Files

Example datasets

Data Analysis



QuantMS Hands-On for PXD04621

- [slides](#)

Open GitHub codespace

Use the following link to open a GitHub codespace with most of the required software installed:

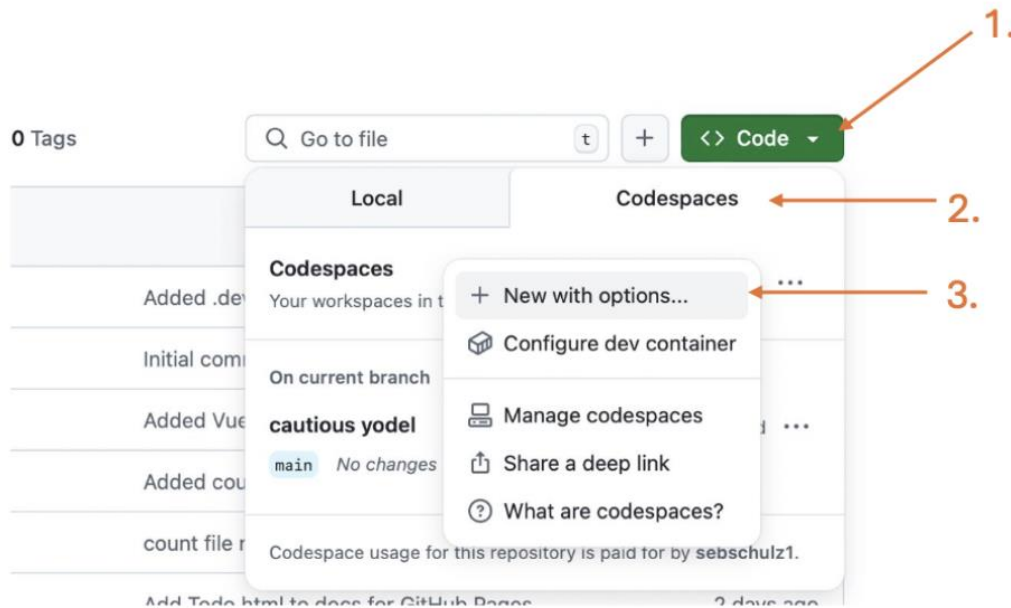
⚠ If you do it manually, make sure to select the bigger machine with 4 cores and 16GB RAM



Open in GitHub Codespaces

Before the break please create your codespace!

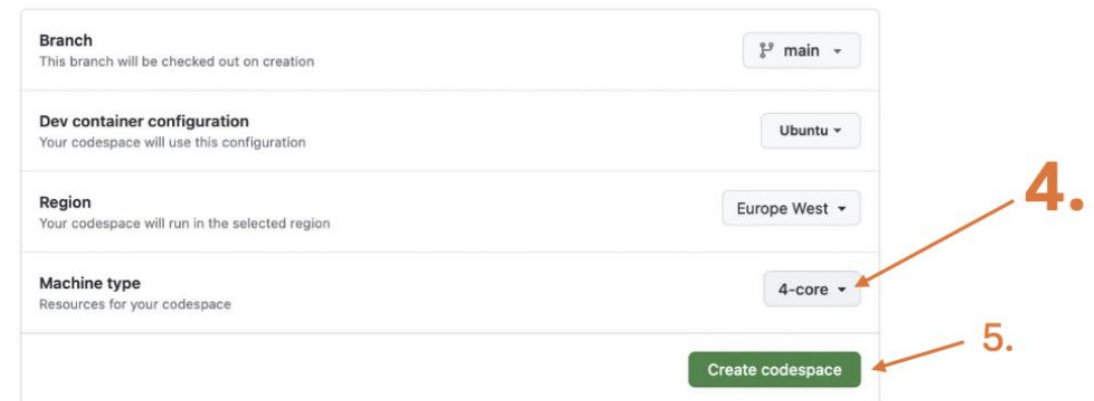
1. Log into your github account
2. Open the training repo: https://github.com/biosustain/dsp_proteomics_training
3. Create codespace with **4 cores** which we will use to run the code
4. Creating the codespace will take about 30 mins



!!! IMPORTANT !!!

Select 4-cores for machine type before creating the codespace

Create codespace for
biosustain/dsp_transcriptomics_training



Let's see this in practice after a break!

Thanks a lot for your attention and
please feel free to ask more question
to me or my team 😊

